

Strongly anharmonic systems with CRYSTAL

M. Calandra



<https://mattheory.physics.unitn.it>



www.sscha.eu



<https://the-epiq-team.gitlab.io/epiq-site/>



European Research Council
Established by the European Commission

OUTLINE

Introduction to the method:



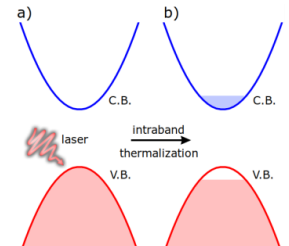
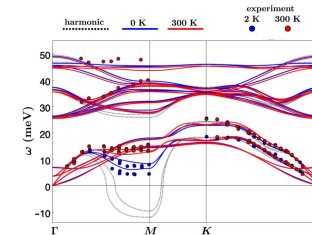
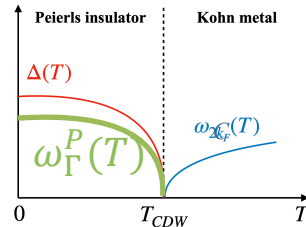
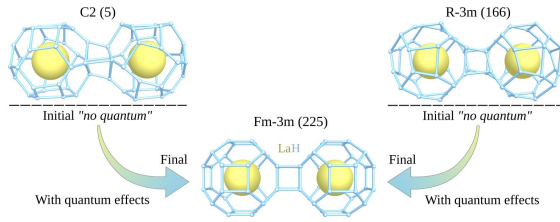
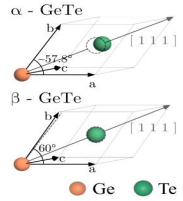
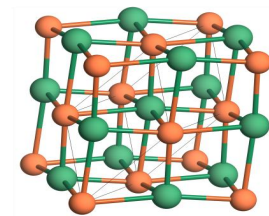
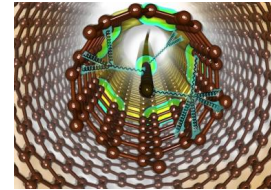
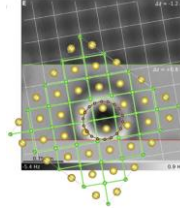
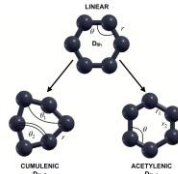
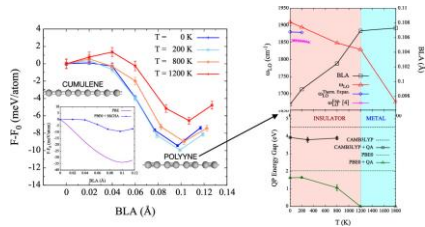
www.sscha.eu

+ Any total energy
and force engine



But not only.....

Some applications depending on the mood of the speaker:



SCHA – Anharmonic effects beyond perturbation theory

SCHA=Self-Consistent Harmonic Approximation

Or how to find the best harmonic approximation for a quantum anharmonic free energy.

LI. *A New Treatment of Anharmonicity in Lattice Thermodynamics : I*

By D. J. HOOTON



D. J. Hooton, Philos. Mag. 46, 522 (1955)

✓ Non perturbative approach

✗ Expensive: we need third and fourth order derivatives of total Energy.

See I. Errea, B. Rousseau and A. Bergara, Phys. Rev. Lett. **106**, 165501 (2011)

How to implement the SCHA in an efficient way ?

SSCHA – Quantum Anharmonicity beyond perturbation theory

SSCHA=Stochastic Self-Consistent Harmonic Approximation

Anharmonic hamiltonian: $H = T + V$

The free energy is:

$$F_H = \text{tr}[\rho_H H] + \frac{1}{\beta} \text{tr}[\rho_H \ln \rho_H] = -\frac{1}{\beta} \ln Z_H$$

Density
matrix
↓

Difficult to calculate density matrix as V includes n^{th} order anharmonicity.

SSCHA – Quantum Anharmonicity beyond perturbation theory

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Density matrix
↓

Difficult to calculate density matrix as V includes n^{th} order anharmonicity.

Consider a trial Hamiltonian $\mathcal{H} = T + \mathcal{V}$ and define the quantity

$$\mathcal{F}_H[\mathcal{H}] = \text{tr}[\rho_{\mathcal{H}} H] + \frac{1}{\beta} \text{tr}[\rho_{\mathcal{H}} \ln \rho_{\mathcal{H}}]$$

« Trial » potential

then

$$F_H \leq \mathcal{F}_H[\mathcal{H}]$$

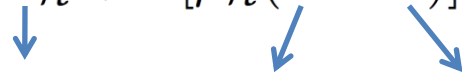
Gibbs-Bogolugov Inequality

SSCHA – Quantum Anharmonicity beyond perturbation theory

$$\mathcal{F}_H[\mathcal{H}] = \text{tr}[\rho_{\mathcal{H}} H] + \frac{1}{\beta} \text{tr}[\rho_{\mathcal{H}} \ln \rho_{\mathcal{H}}]$$

Summing and subtracting $\text{tr}[\rho_{\mathcal{H}} \mathcal{H}]$ on the r.h.s. and making the trace explicit we get:

$$\mathcal{F}_H[\mathcal{H}] = F_{\mathcal{H}} + \text{tr}[\rho_{\mathcal{H}} (V - \mathcal{V})]$$


« Trial » Free energy Real potential « Trial » potential

We evaluate the trace in the position basis:

$$F_H \leq \mathcal{F}_H[\mathcal{H}] = F_{\mathcal{H}} + \int d\mathbf{R} [V(\mathbf{R}) - \mathcal{V}(\mathbf{R})] \rho_{\mathcal{H}}(\mathbf{R}).$$

$$\rho_{\mathcal{H}}(\mathbf{R}) = \langle \mathbf{R} | e^{-\beta \mathcal{H}} | \mathbf{R} \rangle / Z_{\mathcal{H}} \quad \text{« Trial » partition function}$$

What is the advantage of all this ?

SSCHA – Quantum Anharmonicity beyond perturbation theory

Choose the density matrix as Gaussian.

- ☑ $\rho_{\mathcal{H}}(\mathbf{R})$ is Gaussian
- ☑ The « Trial » free energy is analytical ($\rightarrow$ Form of a Harmonic free energy)

$$F_H \leq \mathcal{F}_H[\mathcal{H}] = \underbrace{F_{\mathcal{H}}}_{\text{« Trial » Free energy}} + \int d\mathbf{R} \underbrace{[V(\mathbf{R}) - \mathcal{V}(\mathbf{R})]}_{\substack{\text{Real} \\ \text{potential}}} \underbrace{\rho_{\mathcal{H}}(\mathbf{R})}_{\text{Harmonic Potential}}$$

The integral on the right hand side is evaluated with a Monte Carlo statistical approach:

$$\int d\mathbf{R} O(\mathbf{R}) \rho_{\mathcal{H}}(\mathbf{R}) \simeq \frac{1}{N_c} \sum_{I=1}^{N_c} O(\mathbf{R}_I) \equiv \langle O \rangle$$

N_c = number of ionic configurations distributed as $\rho_{\mathcal{H}}(\mathbf{R})$

Exact for an infinite number of N_c

IMPORTANT: This is the quantum statistical average!

SSCHA – Mapping to effective harmonic Hamiltonian

One-to-one mapping

$$\rho_{\mathcal{H}}(\mathbf{R}) \longleftrightarrow \mathcal{H}_{\mathcal{R},\Phi} = K + \frac{1}{2} \sum_{ab} (R_a - \mathcal{R}_a) \Phi_{ab} (R_b - \mathcal{R}_b)$$

$\rho_{\mathcal{H}}(\mathbf{R})$ is Gaussian

We see that the trial density matrix $\rho_{\mathcal{H}}(\mathbf{R})$ will depend on $\rho_{\mathcal{R},\Phi}$

We can try to minimize the free energy with respect to $\mathcal{R}$ and Φ

In the absence of symmetry these are $(3 \times N_a) \times (3 \times N_a - 1) / 2 + 3 \times N_a$ parameters

$\mathcal{R}$ are the so-called **CENTROIDS**

Φ is the so-called **Auxiliary force constant matrix**

SSCHA – In practice

1. In a supercell, generate gaussian random ionic displacements distributed as $\rho_{\mathcal{H}}(\mathbf{R})$
2. Evaluate from Born-oppenheimer (ab initio) total energies:

$$\mathcal{F}_H[\mathcal{H}] = F_H + \frac{1}{N_c} \sum_{l=1}^{N_c} [V(\mathbf{R}_l) - \mathcal{V}(\mathbf{R}_l)] \rho_{\mathcal{H}}(\mathbf{R}_l)$$

3. From ab initio forces, $\mathbf{f}(\mathbf{R}_l)$, evaluate the gradient of the free energy

$$\nabla \mathcal{F}_H[\mathcal{H}] = \frac{1}{N_c} \sum_{l=1}^{N_c} \mathcal{O}_{\mathcal{H}}[f(\mathbf{R}_l)] \rho_{\mathcal{H}}(\mathbf{R}_l)$$

4. Minimize free energy with conjugate gradient to get the best trial free energy

$$F_H \leq \mathcal{F}_H[\mathcal{H}]$$

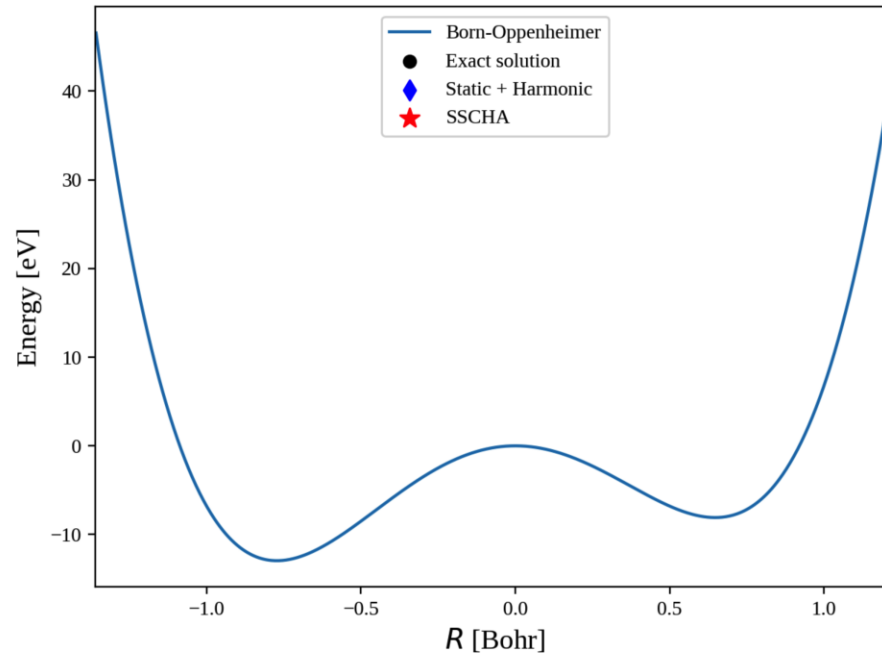
+ reweighting techniques to minimize the number of calls to the ab initio force engine.

Ionic quantum effects included (i.e. the ions have a gaussian wavefunction)

A simple one dimensional example

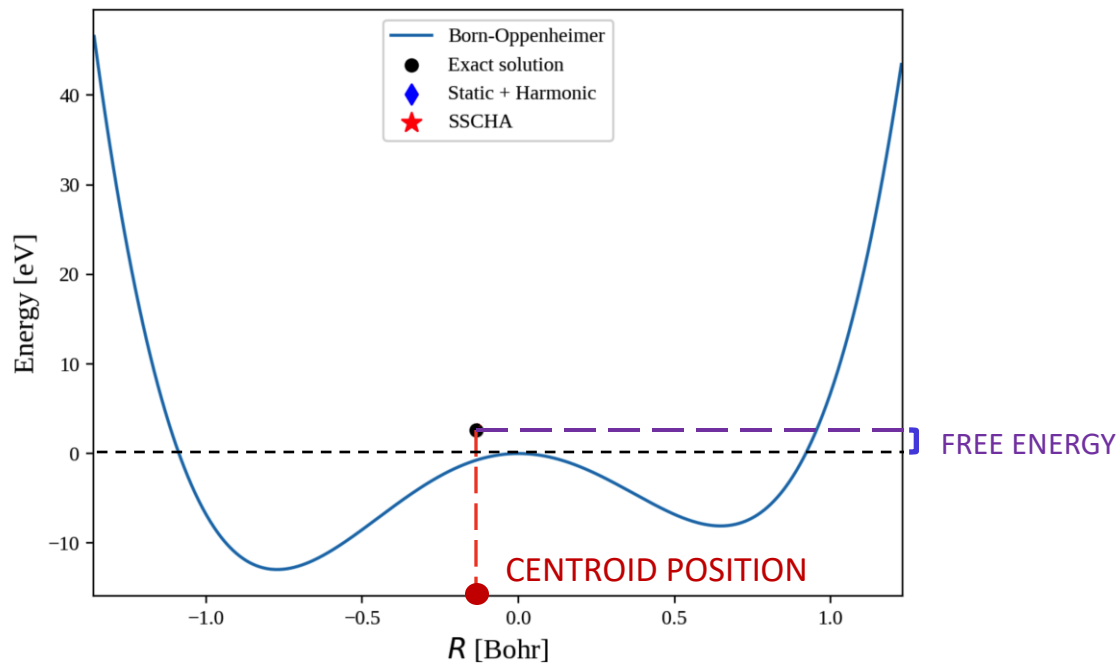
Atomic units

$$V(R) = 3R^4 + \frac{1}{2}R^3 - 3R^2$$



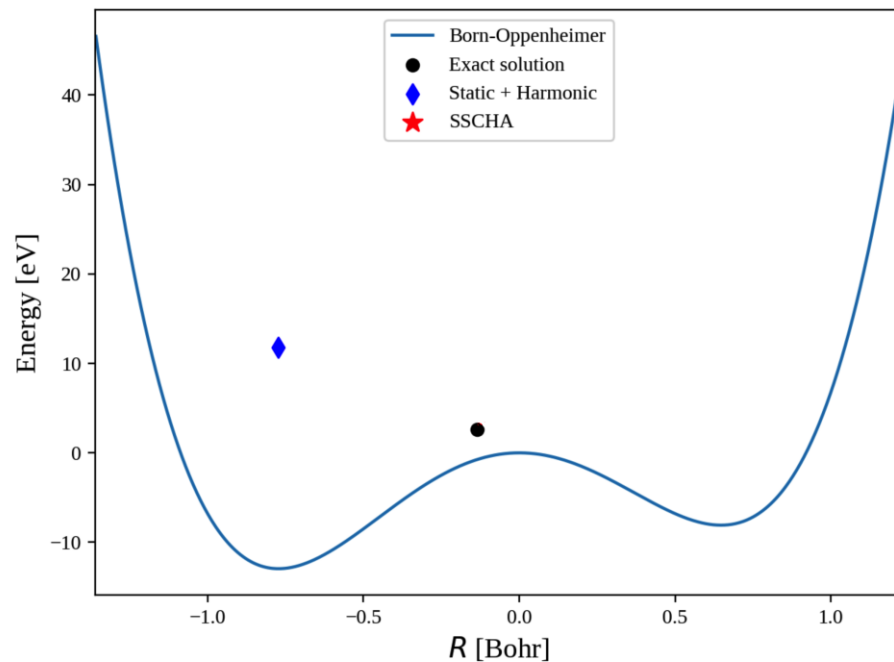
A simple one dimensional example

$$V(R) = 3R^4 + \frac{1}{2}R^3 - 3R^2$$



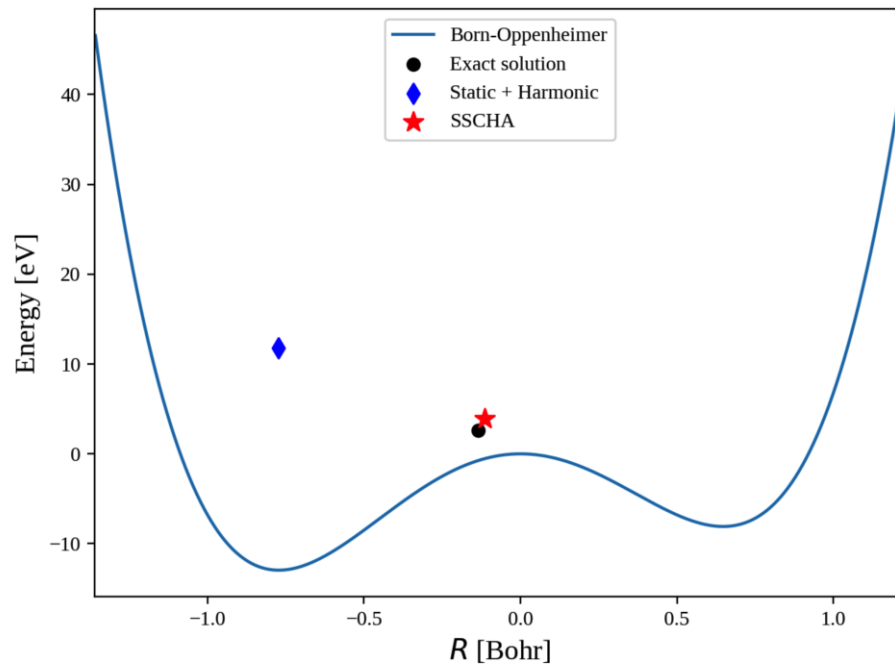
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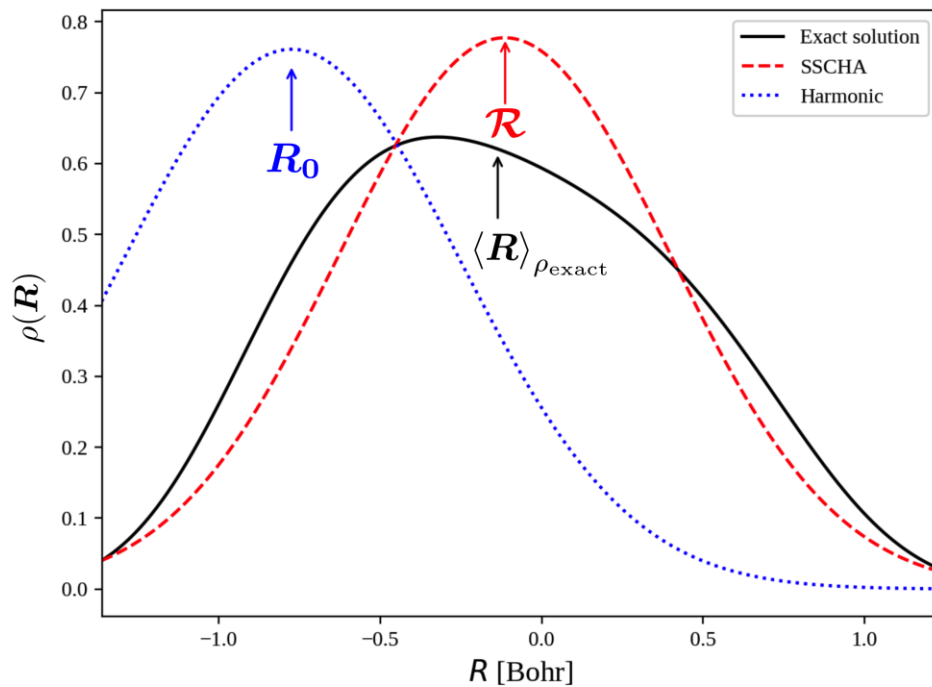
A simple one dimensional example

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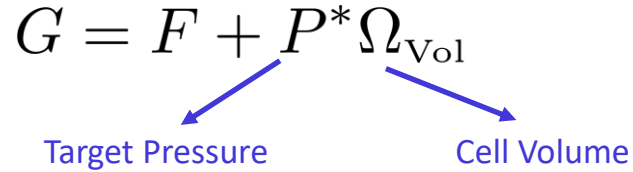


A simple one dimensional example

$$V(R) = 3R^4 + \frac{1}{2}R^3 - 3R^2$$



SSCHA – Free-energy minimization at finite Pressure

$$G = F + P^* \Omega_{\text{Vol}}$$


Target Pressure

Cell Volume

The Gibbs-Bogolubov inequality gives:

$$G \leq \mathcal{G}[\mathcal{R}, \Phi] = \mathcal{F}[\mathcal{R}, \Phi] + P^* \Omega_{\text{Vol}}$$

Thus, we can minimize the Gibbs free-energy and obtain the quantum pressure.

SSCHA – Phonon frequencies

From the auxiliary force constant matrix we could obtain SSCHA phonon frequencies

$$\mathcal{H}_{\mathcal{R},\Phi} = K + \frac{1}{2} \sum_{ab} (R_a - \mathcal{R}_a) \Phi_{ab} (R_b - \mathcal{R}_b)$$

However, these are **not physically relevant quasi-particles**.

Conceptually similar to DFT quasiparticles, i.e. they are not physically relevant (in principle)

Moreover SSCHA quasi particles are always positive definite, structural instabilities ?

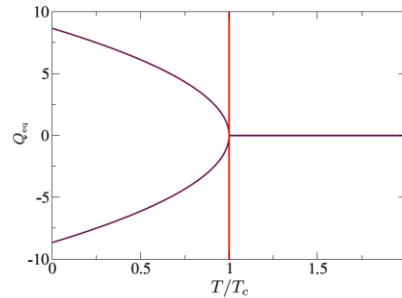
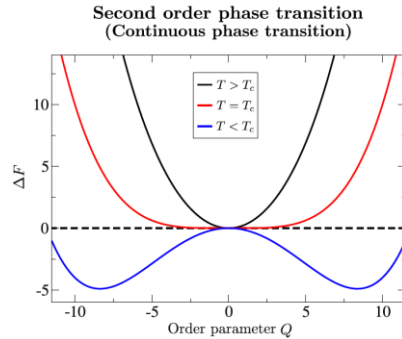
How can we obtain physically relevant quasiparticles ?

SSCHA – Second order structural instabilities

The instability is determined by the Free energy Hessian:

$$\Delta F = \frac{\partial F}{\partial Q_i \partial Q_j}$$

Where **Q** is the order parameter (atomic displacement)



Within the SSCHA, we developed an exact analytic expression for ΔF as average of a function of total energy and forces over Displaced ionic configurations.

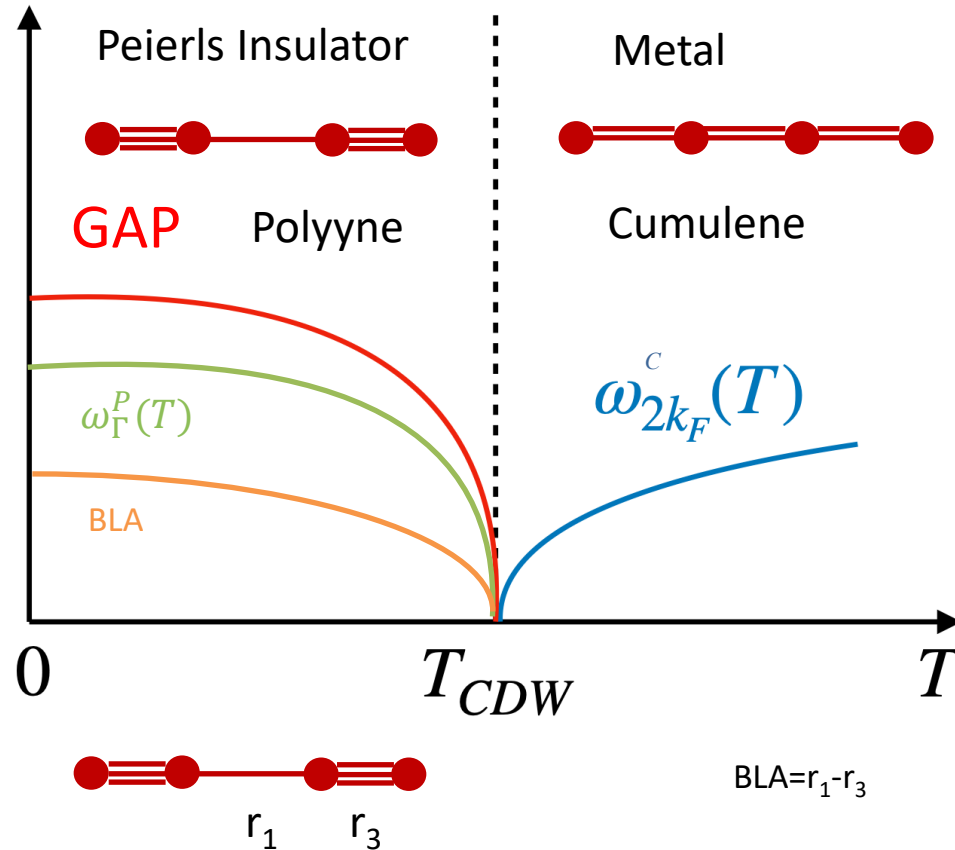
One shot calculation of energy and of its Hessian

Phonon spectral weight and quasiparticles can be obtained from the Hessian

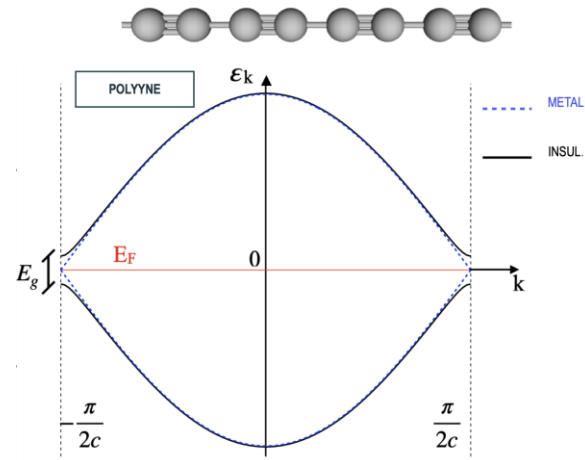
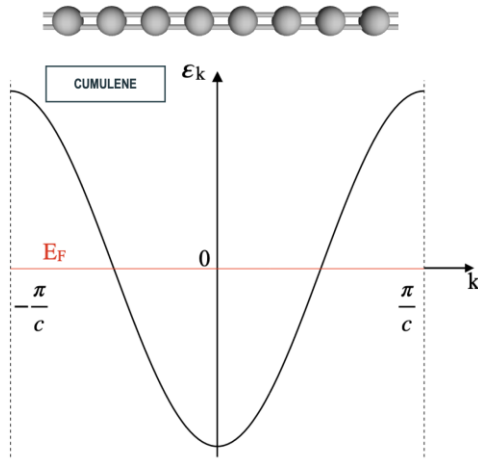
Dominant Role of Quantum Anharmonicity in the Stability and Optical Properties of Infinite Linear Acetylenic Carbon Chains

D. Romanin, L. Monacelli, R. Bianco, I. Errea, F. Mauri, and M. Calandra, J. Phys. Chem. Lett. **12**, 10339 (2021)

1D Carbon chain the prototypical realization of a Peierls distortion (?)

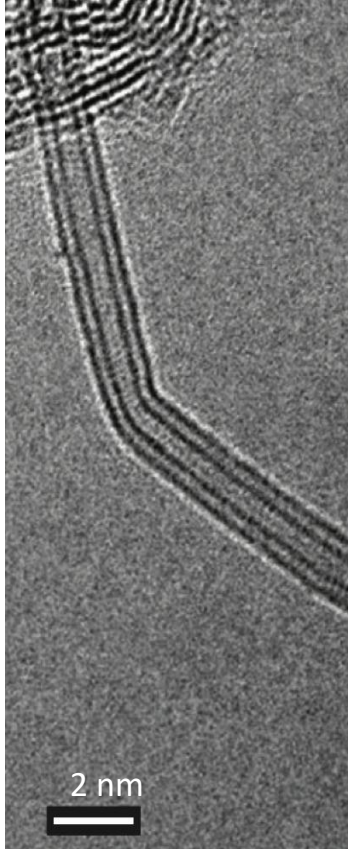


Peierls mechanism (small displacement limit)

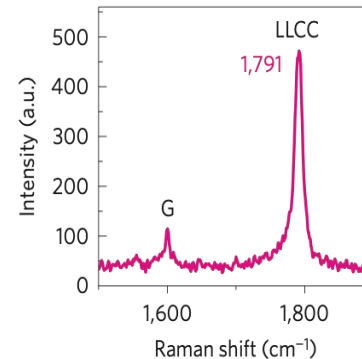
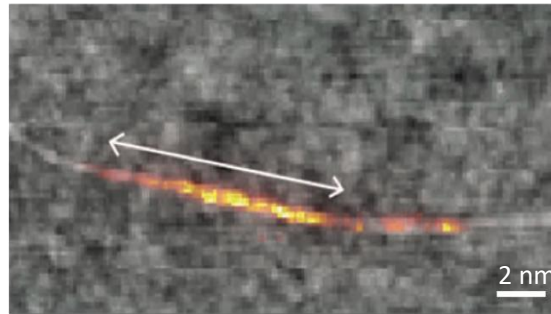


Energy gain due to band-gap opening should be order of the gap.

Physical realization of carbyne



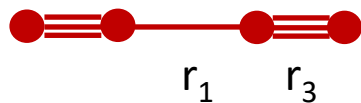
- Cumulene has never been observed;
- Polyyne can be obtained by end-capping (free-standing) and inside carbon nanotubes;
- Ultralong chain (~ 6000 atoms) can be encapsulated in DWCNTs;
- In all of these situations, a distorted phase is always observed.



L. Shi et al., *Nature Materials* **15**, 634–639 (2016)

Exchange and correlation effects

										Bandgap	
method	n	MF	X	C	QA	$2c$ (Å)	r_1 (Å)	r_3 (Å)	BLA (Å)	ΔE (eV)	ω_{LO} (cm ⁻¹)
This work											
DFT-PBE0	PBC	•	•			2.534	1.3156	1.2183	0.0973	2.03	1986.57
DFT-CAMB3LYP	PBC	•	•			2.526	1.3286	1.1973	0.1313	4.39	2277.22
Literature											
DMC ⁴⁵	PBC		•	•		2.5817(9)	1.359(2)	1.223(2)	0.136(2)	3.4(1)	2084(5)
RPA (vacuum) ⁴⁶	PBC	•	•	•		2.590	1.36	1.23	0.129	—	2000
HF ⁴²	72	•	•			—	—	—	0.183	8.500	—
MP2 ⁴²	40		•	•		—	—	—	0.060	5.541	—
MP2 ⁴¹	PBC		•	•		2.554	1.337	1.217	0.120	—	—
CCSD(T) ⁴⁸	36		•	•		—	—	—	0.125	—	2090
CCSD(T) ⁴¹	PBC		•	•		2.565	1.358	1.207	0.151	—	—
DFT-LDA ⁴¹	PBC	•				2.532	1.286	1.246	0.040	—	—
DFT-PBE ⁴⁵	PBC	•				2.565	1.300	1.265	0.035	—	—
DFT-HSE06 ⁴⁵	PBC	•	•			2.56	1.323	1.237	0.086	—	—
DFT-B3LYP ⁴²	72	•	•			—	—	—	0.088	1.487	—

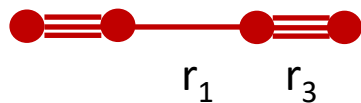


$$\text{BLA} = r_1 - r_3$$

All functionals give polyne as the most stable structure

Exchange and correlation effects

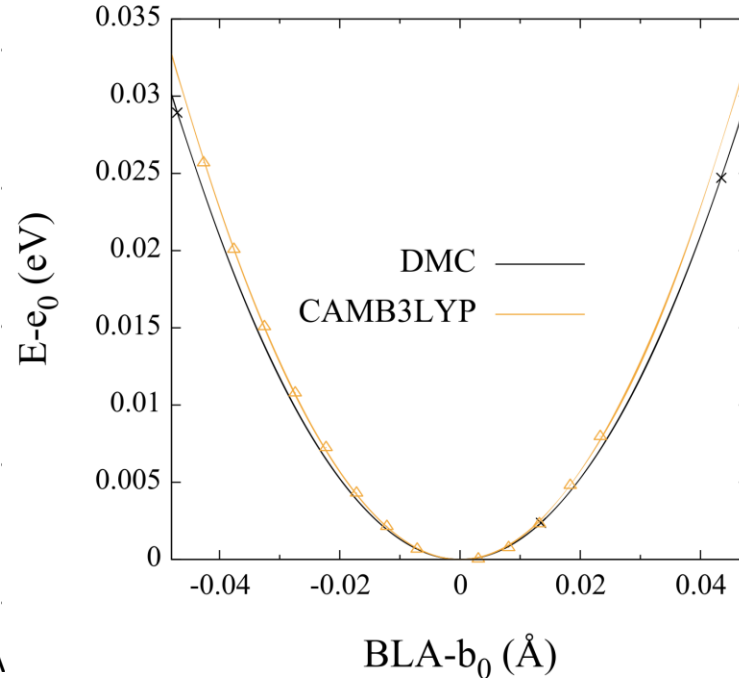
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$$\text{BLA} = r_1 - r_3$$

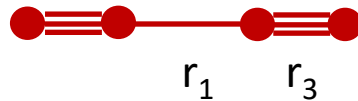
The CAMB3LYP is the most accurate in vacuum

Accuracy of the CAMB3LYP : energy versus BLA



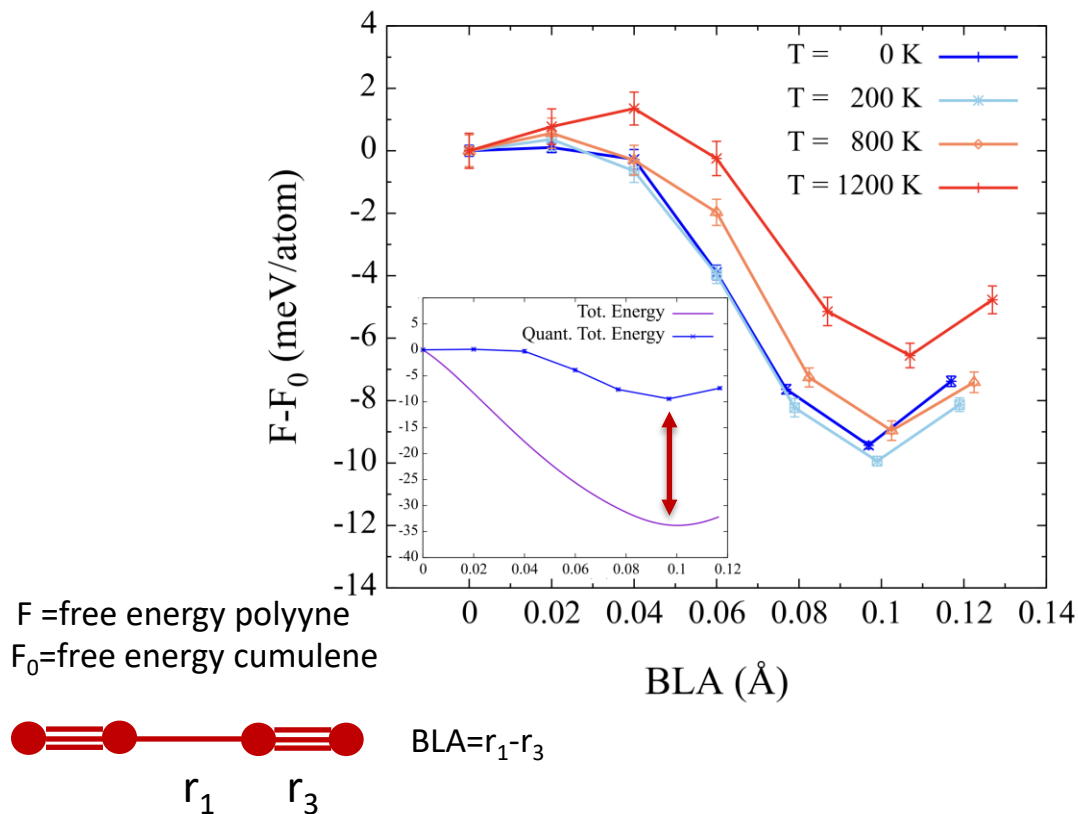
b_0 = optimal BLA

e_0 = total energy at optimal BLA



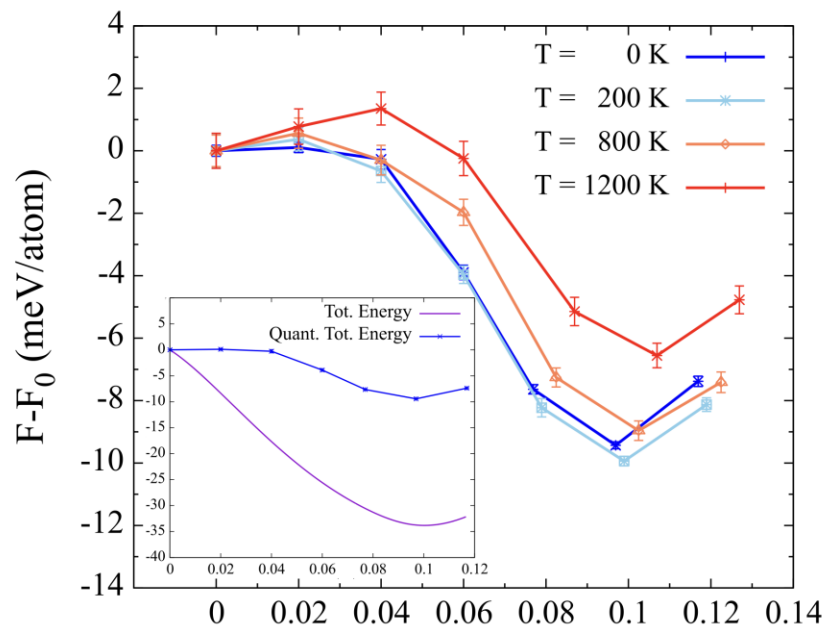
$$BLA = r_1 - r_3$$

Quantum anharmonic free energy via the SSCHA@CRYSTAL



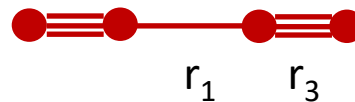
Colossal suppression of the energy gain by the distortion (71%) !

Quantum anharmonic free energy via the SSCHA@CRYSTAL



F = free energy polyne
 F_0 = free energy cumulene

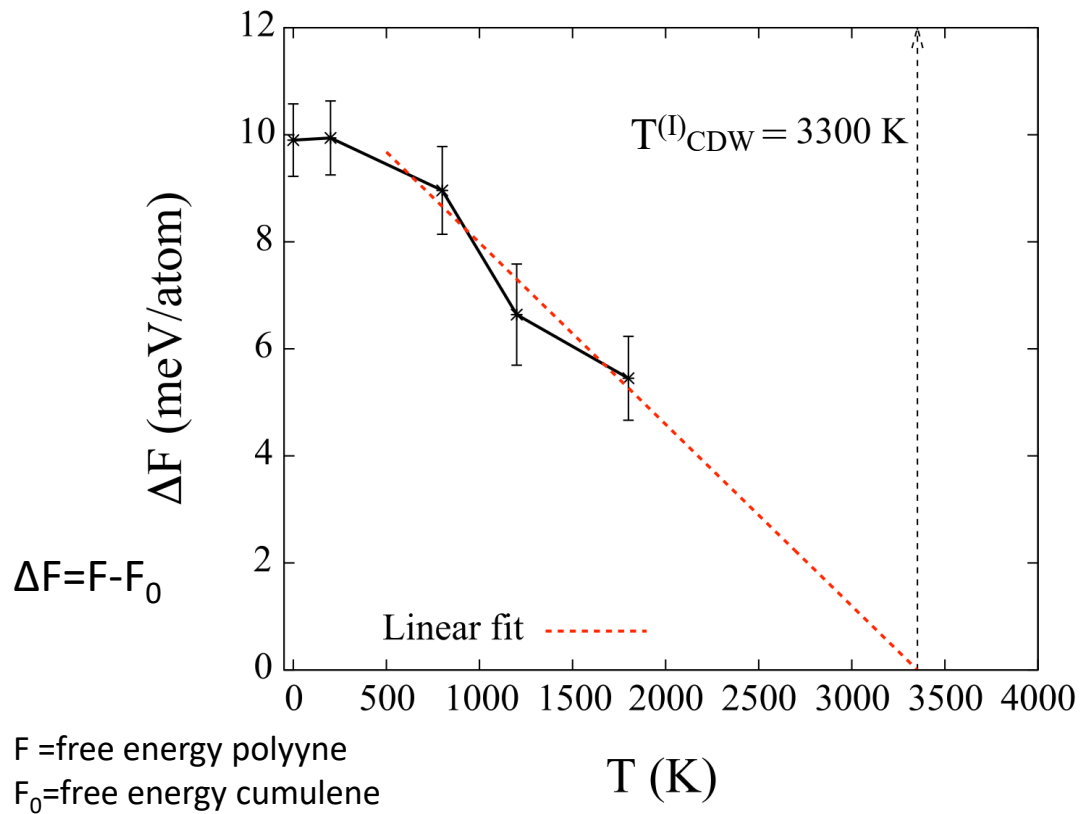
BLA (Å)



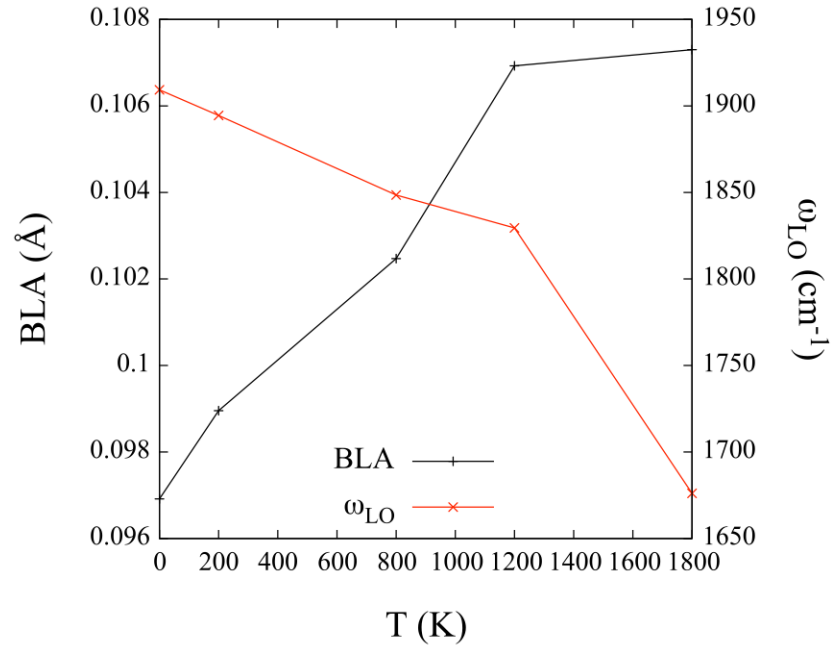
BLA = $r_1 - r_3$

The polyne phase stable even at very high temperature.

Estimated polyynes to cumulene transition at 3500 K

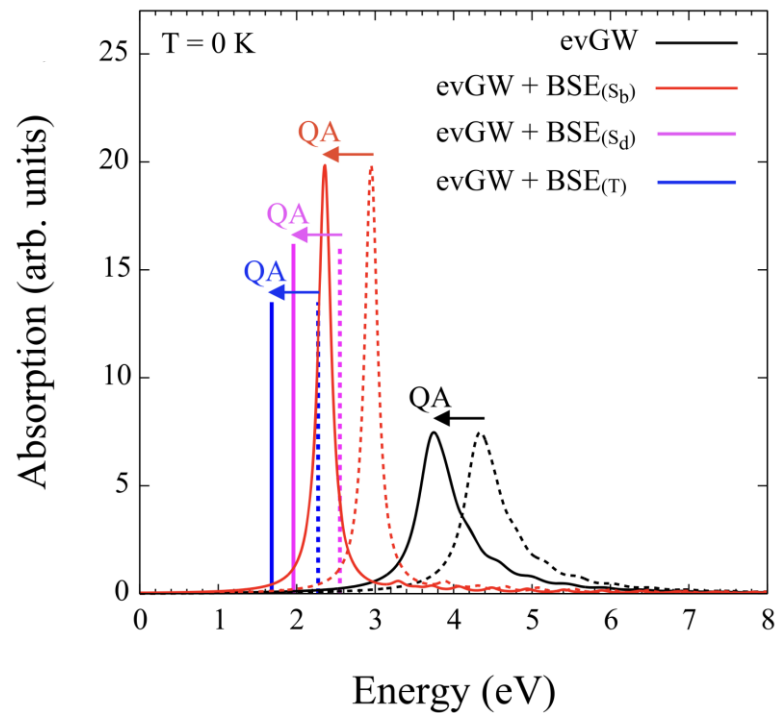


Even more surprising : the BLA increases with increasing T !



In strong contrast with the Peierls picture.

Huge effects of quantum anharmonicity on optical absorption



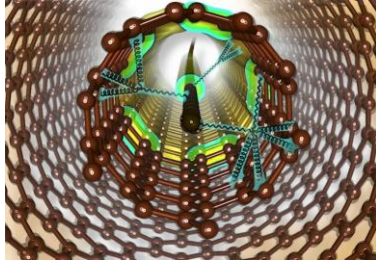
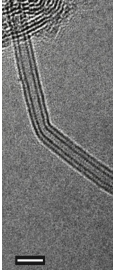
Carbyne - Conclusion

- .1. Quantum-anharmonicity reduces the energy gain of the polyyne phase with respect to the cumulene one by 71%.
- .2. The transition to the cumulene phase happens at extremely high temperatures (i.e. cumulene never stable in the absence of strain).
- .3. The bond length alternation *increases* with *increasing* temperature (at odd with Peierls picture)
- .4. The optical properties are strongly affected by quantum anharmonic effects

After solving the problem of isolated carbyne we thought we could face the problem of carbyne in a nanotube.....

Carbyne Inside a nanotube (after 4 years of work!)

Emil Parth *et al.* Nature Communications 19, 4797 (2025)

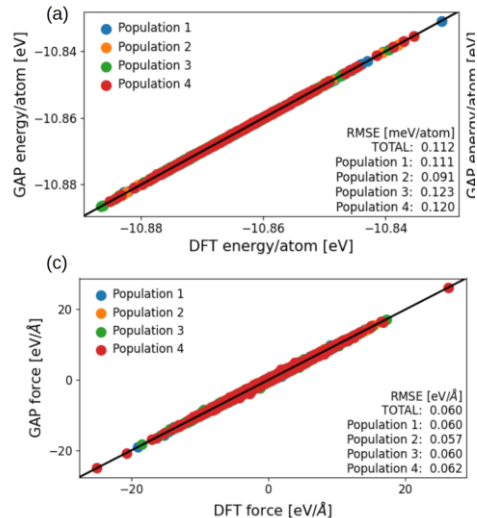


Extremely difficult problem:

- 1 The system is actually a double wall but we only consider the effects of the internal one.
- 2 The non perfect matching between carbyne and the nanotube lattice parameters introduces a substantial strain.
- 3 The screening depends on the nanotube surrounding carbene, (i.e. different range and magnitude of exchange for each radius.....)
- 4 Very large anharmonic effects and strongly dependent on screening!

Step 1: we fitted a (very accurate) machine learning gap/mace potential

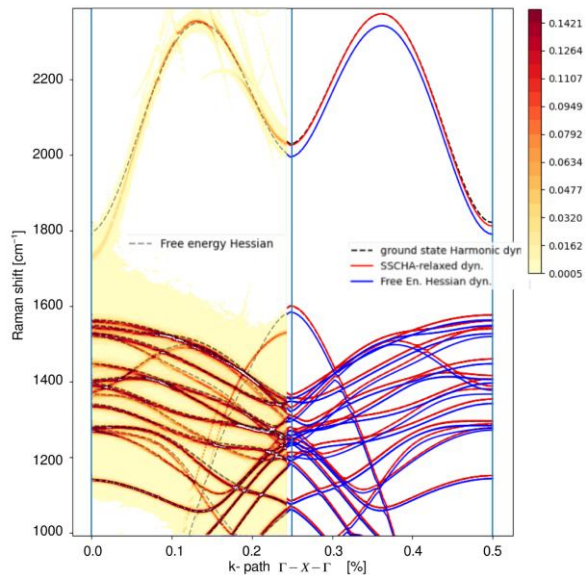
- 1 We considered a [5,5] metallic nanotube
- 2 We slightly stretch it to compensate for the carbyne/nanotube lattice mismatching
- 3 HSE with parameters 0.3 exchange fraction and range separation $0.07 \text{ \AA}^{-1}$
(needless to say that LDA/GGA or B3LYP completely fail – more than 200 cm^{-1} error)



Extremely accurate energy and forces

Step 2: we performed SSCHA with the machine learning potential

Anharmonic effects of the order of 100 cm⁻¹ in some cases



We obtained a semiquantitative agreement with experiments.

But (Step 3) we explained double resonant Raman spectra....

P.S. Double resonant Raman is so complicated that it would require several separate lectures....

Theoretical Modeling of Ultrafast Phase Transitions via constrained density functional perturbation theory and SSCHA

S. Mocatti, M. Furci, A. Corradini, K. Holtgrewe, G. Marini, P. Cudazzo and M. Calandra



Giovanni Marini and Matteo Calandra, Phys. Rev. B **104**, 144103 (2021)

S. Mocatti, G. Marini and M.C., J. Phys. Chem. Lett. **14**, 41, 9329 (2023)

M. Furci, G. Marini and M.C. , Phys. Rev. Lett. **132**, 236101 (2024)

H. Padma *et al.*, Nature Materials (2025), 10.1038/s41563-025-02254-2

A. Corradini, G. Marini and M. C. **11**, 151 (2025)

M. Huber *et al.*, Science Advances **10**, ead14481(2024).

K. Holtgrewe, G. Marini and M.C. Nano Letters 2024 24 (42), 13179-13184.

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Giovanni Marini and Matteo Calandra, Phys. Rev. B **104**, 144103 (2021)

S. Mocatti, G. Marini and M.C., J. Phys. Chem. Lett. **14**, 41, 9329 (2023)

M. Furci, G. Marini and M.C. , Phys. Rev. Lett. **132**, 236101 (2024)

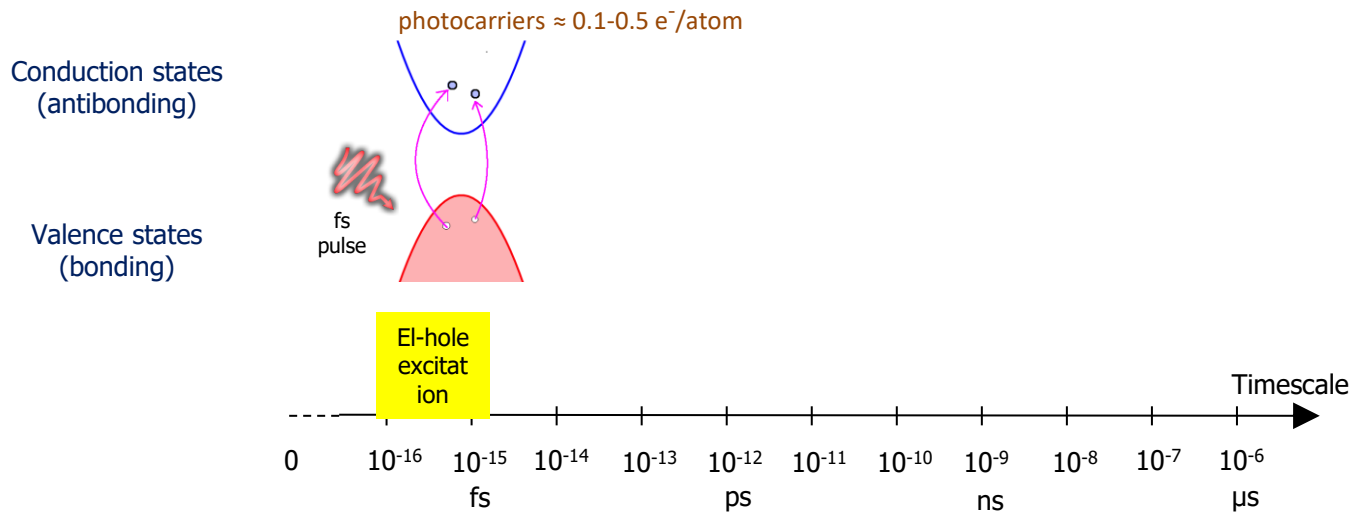
H. Padma *et al.*, Nature Materials (2025), 10.1038/s41563-025-02254-2

A. Corradini, G. Marini and M. C. **11**, 151 (2025)

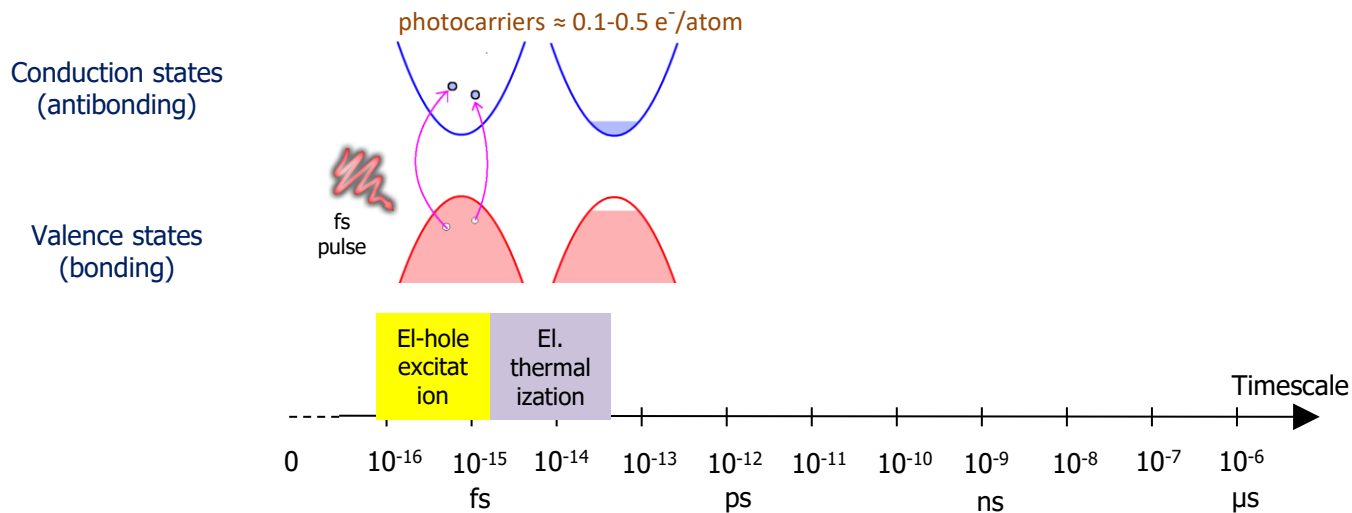
M. Huber *et al.*, Science Advances **10**, ead14481(2024).

K. Holtgrewe, G. Marini and M.C. Nano Letters 2024 24 (42), 13179-13184.

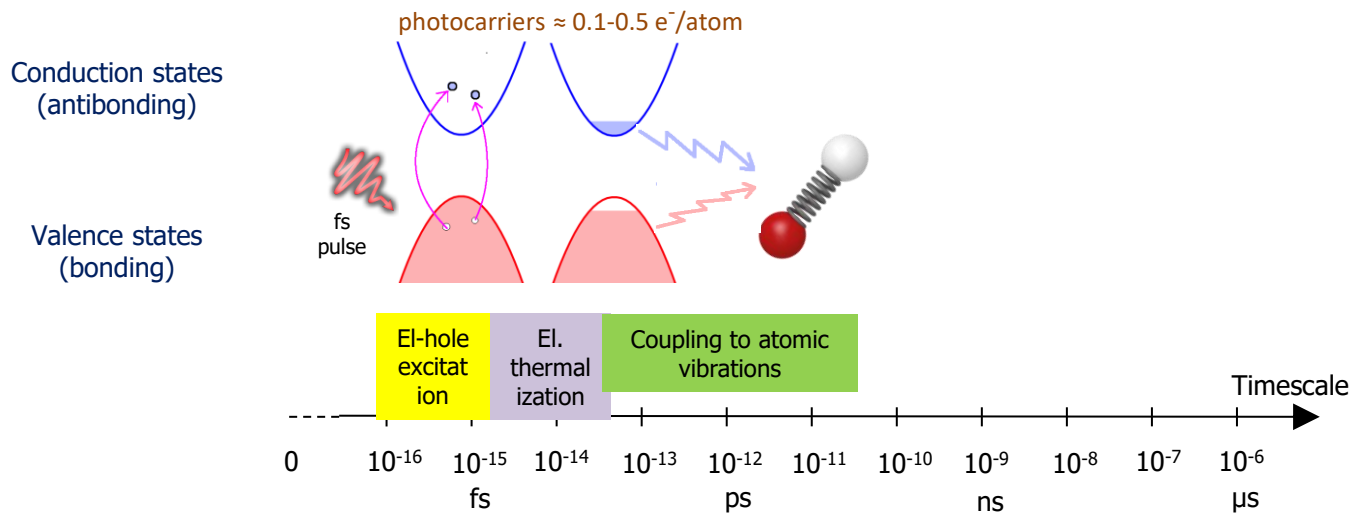
Ultrafast phase transitions in extended systems (see femtochemistry)



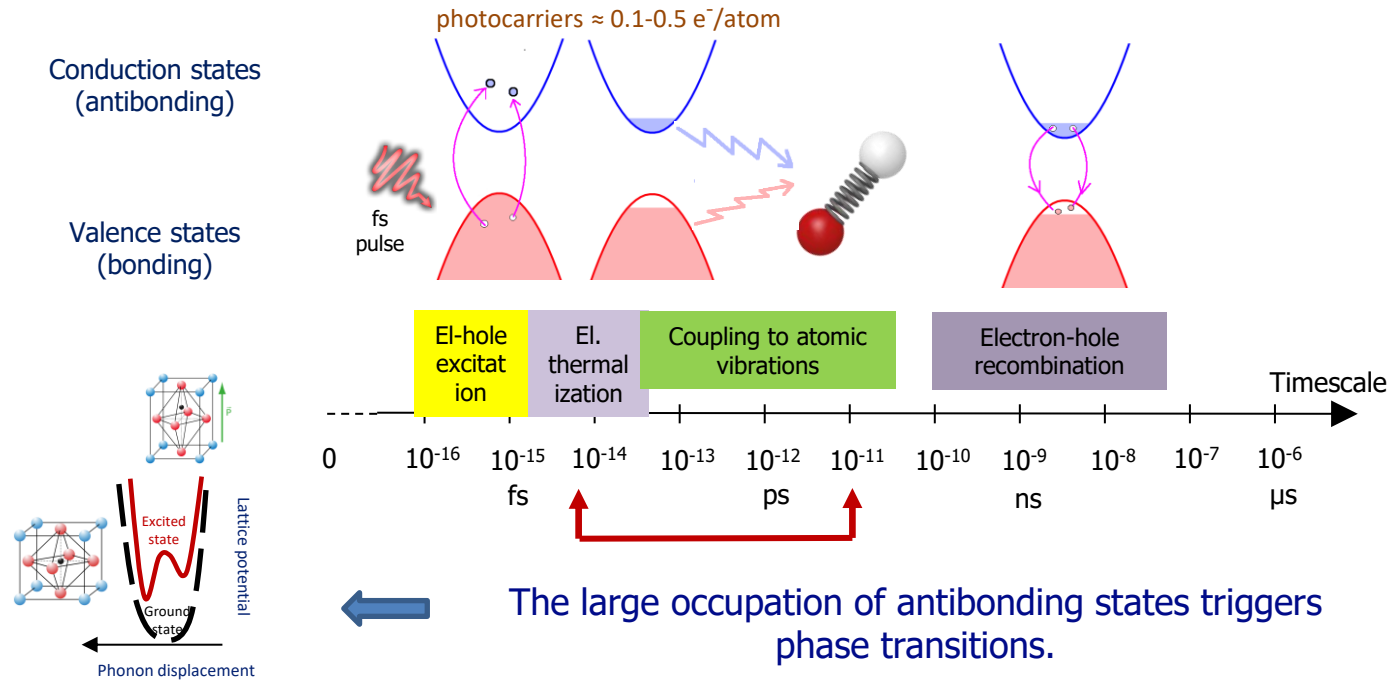
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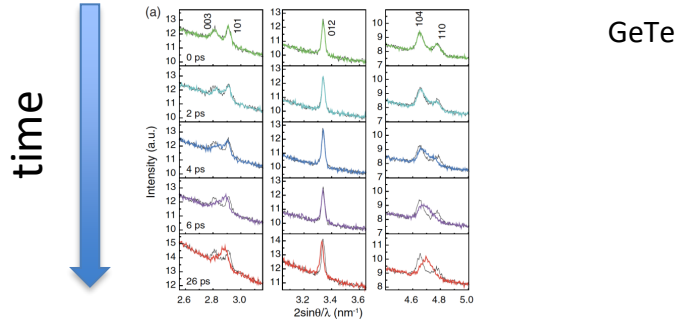
Ultrafast phase transitions in extended systems (see femtochemistry)



We are interested in the photocarriers thermalization and in the phase transitions occurring after electron thermalization.

How do we prove the occurrence of a phase transition ?

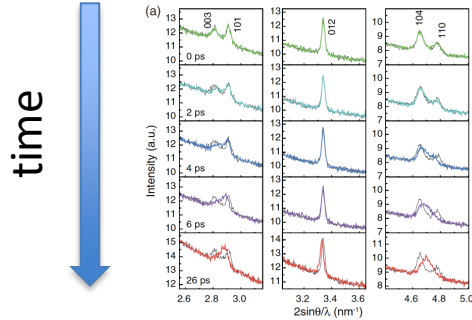
- Ultrafast diffraction via X-ray free electron laser



E. Matsubara *et al.* PRL **117**, 135501 (2016)

How do we prove the occurrence of a phase transition ?

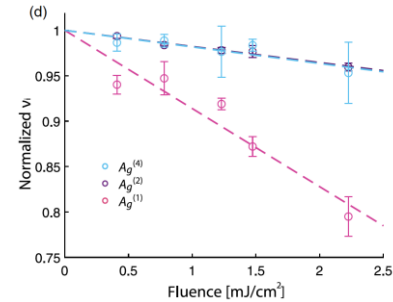
- Ultrafast diffraction via X-ray free electron laser



GeTe

E. Matsubara *et al.* PRL **117**, 135501 (2016)

- Raman active modes dispersive excitation of coherent Phonons (reflectivity oscillations)

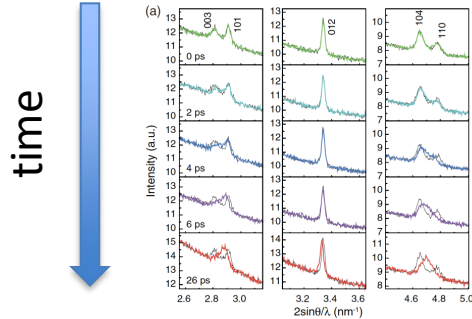


SnSe

Huang *et al.* PRX **12**, 011029 (2022)

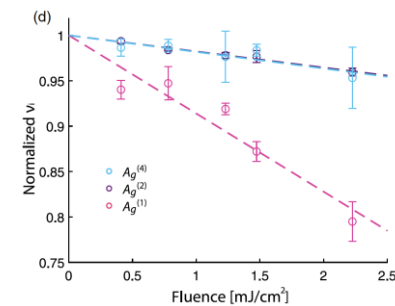
How do we prove the occurrence of a phase transition ?

- Ultrafast diffraction via X-ray free electron laser



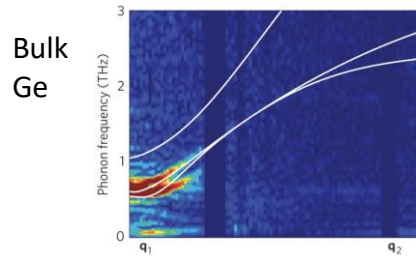
E. Matsubara *et al.* PRL **117**, 135501 (2016)

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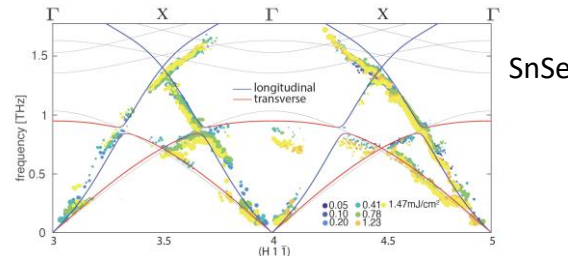


Huang *et al.* PRX **12**, 011029 (2022)

- Phonon dispersion via fs thermal diffuse scattering



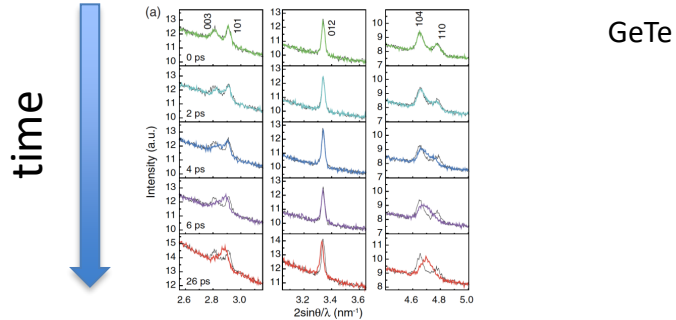
M. Trigo *et al.* Nat. Phys **9**, 790 (2013)



Huang *et al.* arXiv:2301.08955v1

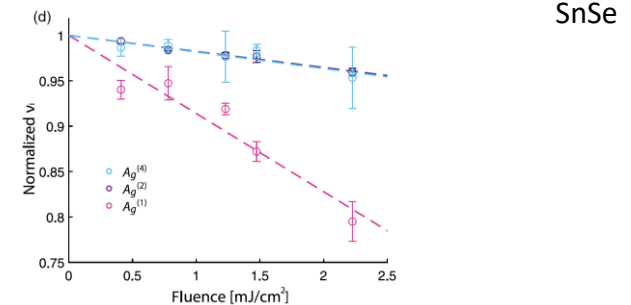
How do we prove the occurrence of a phase transition ?

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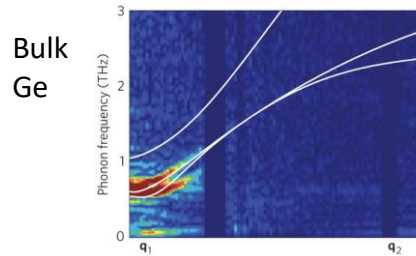
E. Matsubara *et al.* PRL **117**, 135501 (2016)

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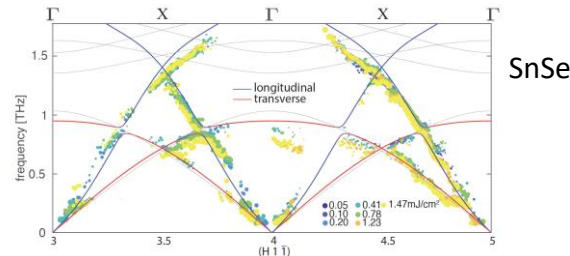


Huang *et al.* PRX **12**, 011029 (2022)

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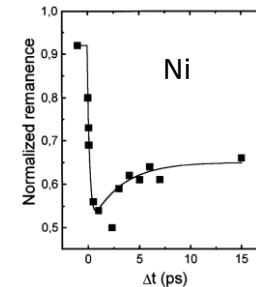


M. Trigo *et al.* Nat. Phys **9**, 790 (2013)



Huang *et al.* arXiv:2301.08955v1

- Magnetism via ultrafast MOKE



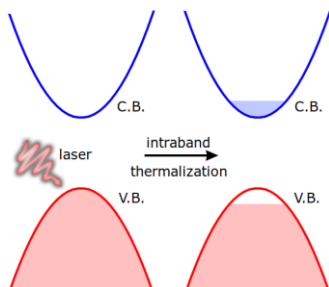
E. Beaurepaire *et al.* PRL **76**, 4250 (1996)

How to efficiently simulate photoexcited phonon and structural dynamics?

Constrained Density functional perturbation theory for photexcited insulators

Giovanni Marini and Matteo Calandra, Phys. Rev. B **104**, 144103 (2021)

- Complete cDFPT framework : two different chemical potentials and electron-hole plasma distribution, no supercells.



We build on top of Refs.:

S. De Gironcoli, Phys. Rev. B **51**, 6773(R) (1995)

Tangney & Fahy, PRL **82**, 4340 (1999), PRB Phys. Rev. B **65**, 054302 (2002)

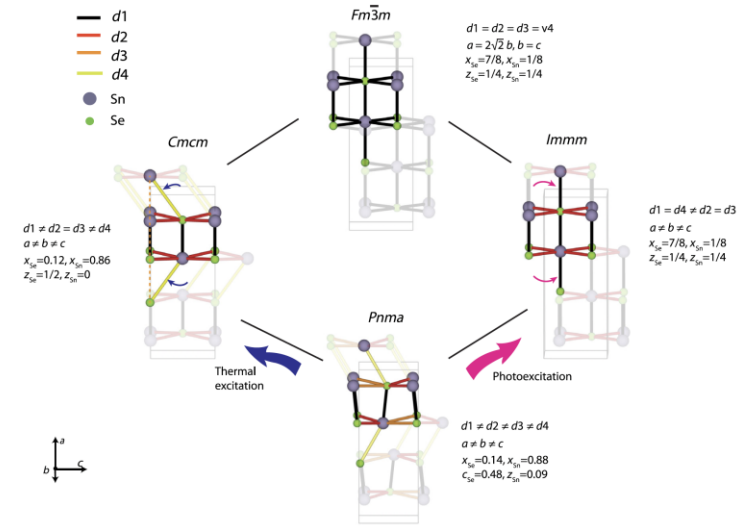
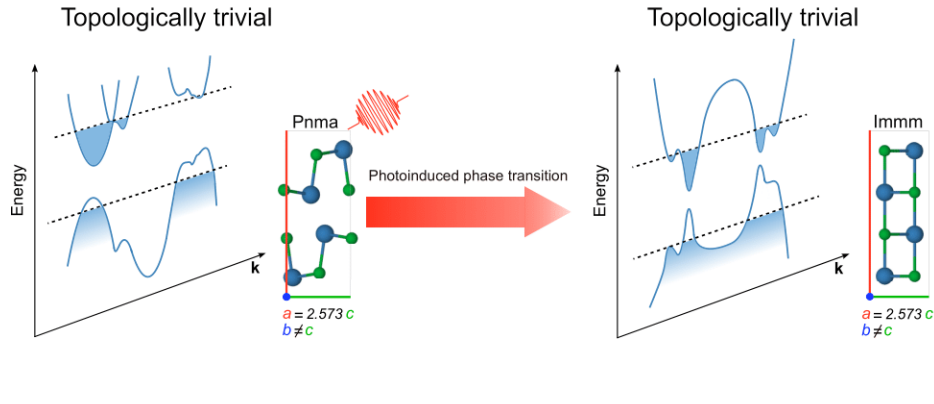
- Complete structural optimization (forces/cell optimization).
- Vibrational properties (phonon dispersion)
- Electron-phonon interaction (including photocarriers-photocarriers screening)
- Coupling with SSCHA, Machine-learning based MD, CPMD (next future), CRYSTAL MD (why not ?).

How does all this framework works ?

Photoexcited SnSe

S. Mocatti, G. Marini and M.C., J. Phys. Chem. Lett. **14**, 41, 9329 (2023)

Non-thermal pathway to the topologically crystalline insulator state in SnSe



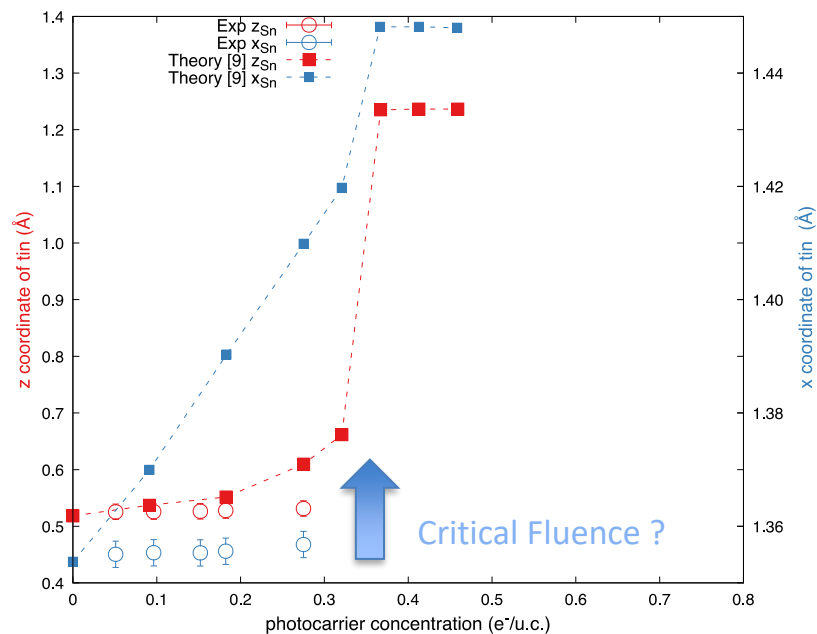
Huang *et al.* PRX 12, 011029

Thermal phase transition : Pnma $\rightarrow$ Cmcm ($T_c=807$ K)

Non-Thermal phase transition : Pnma $\rightarrow$ Imm ?

Critical fluence ? **What happens after recombination ?**

Ultrafast XRD in SnSe as a function of photocarrier concentration

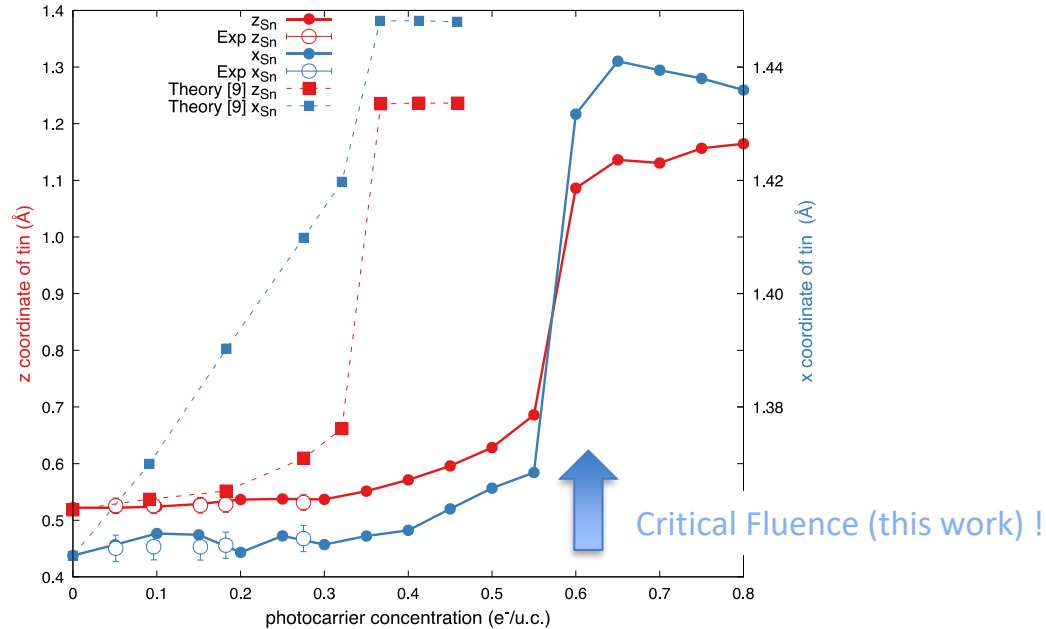


Huang *et al.* PRX 12, 011029

Strong disagreement Theory Experiment.

Incorrect estimate of the critical fluence for the transition.

Ultrafast XRD in SnSe as a function of photocarrier concentration



Huang *et al.* PRX 12, 011029 (2022)

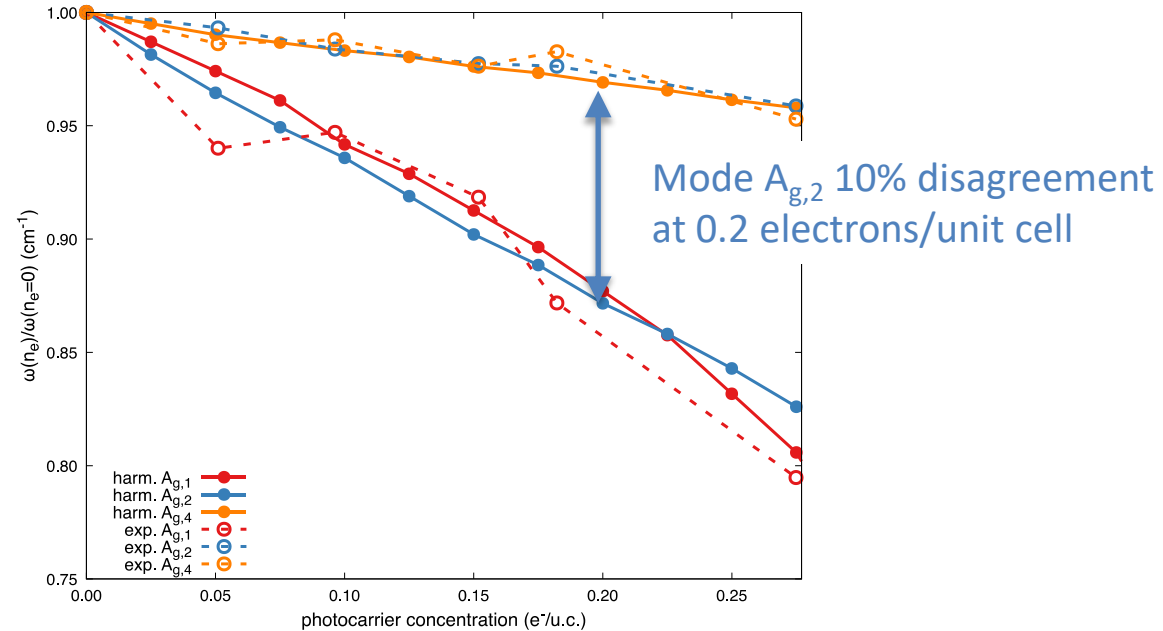
Excellent agreement theory v.s. experiments when the electron-hole plasma is explicitly included.

Experiments in Huang *et al.* still quite far from the critical fluence.

S. Mocatti, G. Marini and M.C., J. Phys. Chem. Lett. 14, 41, 9329 (2023)

G. Marini and M.C., PRB 104, 144103 (2021)

Displacive excitations coherent phonons in SnSe

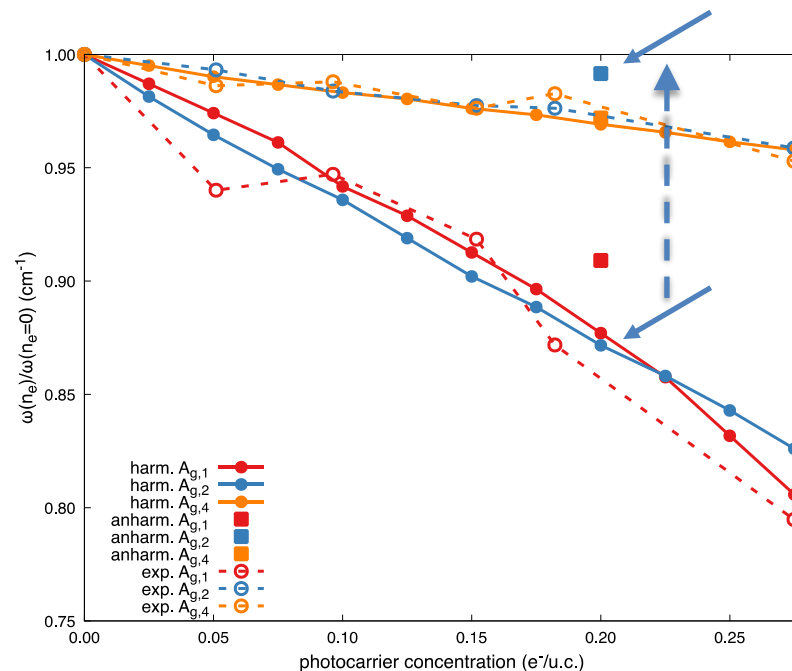


Huang *et al.* PRX 12, 011029 (2022)

Harmonic theory: excellent agreement except for mode A_{g,2}

S. Mocatti, G. Marini and M.C., J. Phys. Chem. Lett. 14, 41, 9329 (2023)

Displacive excitations coherent phonons in SnSe



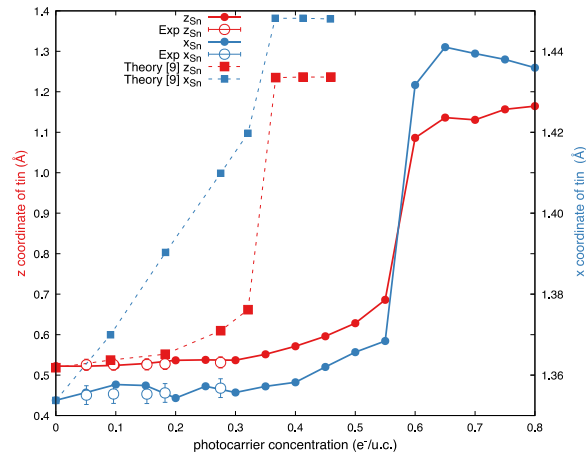
Huang *et al.*
PRX 12, 011029

Crucial role of light-induced quantum anharmonicity !

Error on the quantum free energy curvature of the photoexcited state below 2% !

New experiment with better heat dissipation spot on our prediction !

Predicted critical fluence
 5 mJ/cm^2

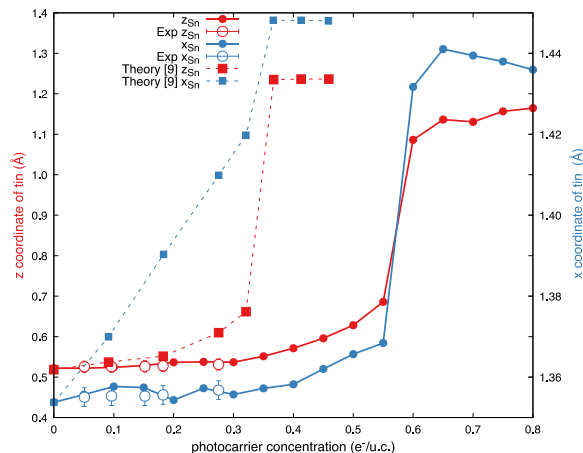


Critical Fluence (this work) !

S. Mocatti, G. Marini and M.C.,
J. Phys. Chem. Lett. 14, 41, 9329 (2023)

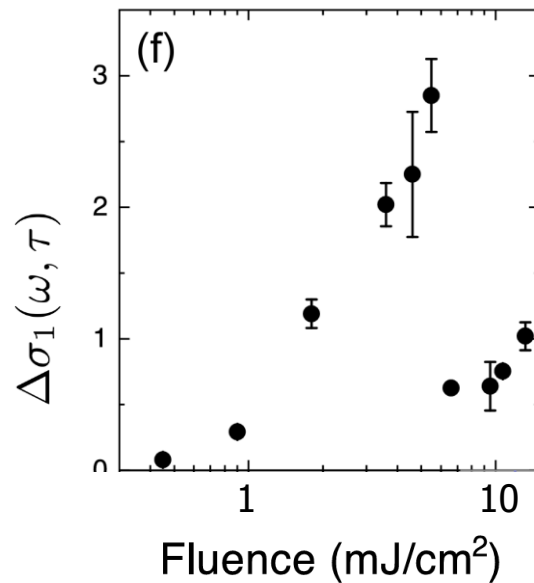
New experiment with better heat dissipation spot on our prediction !

Predicted critical fluence
5 mJ/cm²



Critical Fluence (this work) !

At 6 mJ/cm² sharp drop in THz
differential conductivity



$\omega=10$ THz, $\tau=1$ ps

S. Mocatti, G. Marini and M.C.,
J. Phys. Chem. Lett. 14, 41, 9329 (2023)

ONLY 1 mJ/cm² difference in critical fluence !

Stochastic self-consistent harmonic approximation



www.sscha.eu



<https://the-epiq-team.gitlab.io/epiq-site/>



For scientific updates checkout

<https://mattheory.physics.unitn.it>



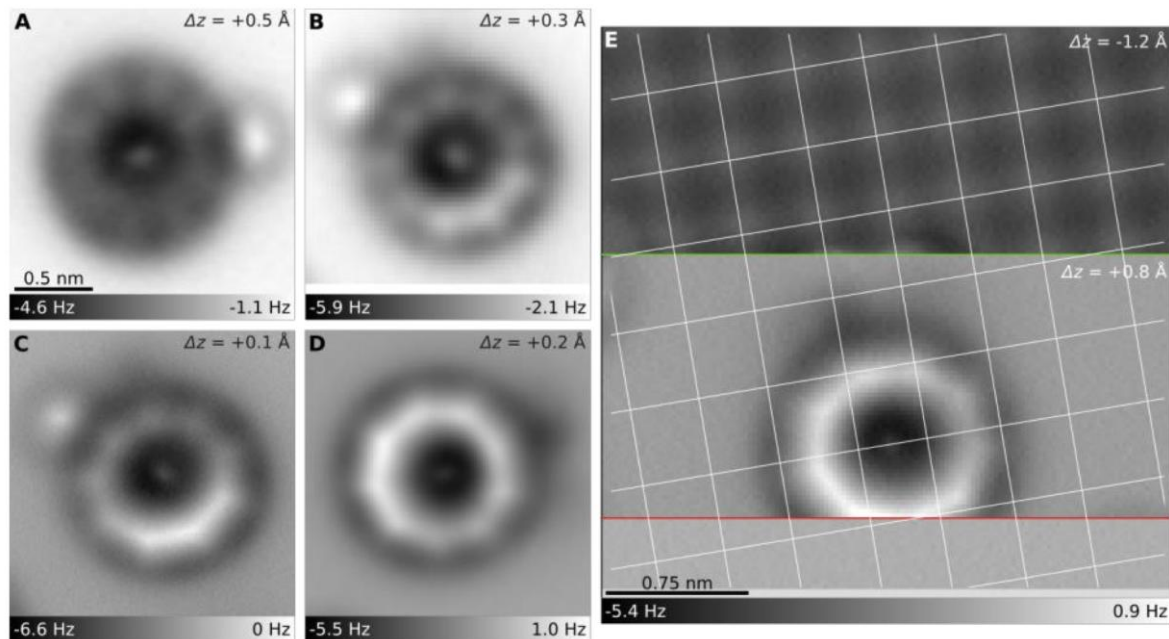
Giant quantum anharmonic effects on the stability, vibrational
and optical properties of cyclo[$4n + 2$]carbon

D. Romanin and M. Calandra, Carbon Trends **9**, 100207 (2022)

An sp-hybridized molecular carbon allotrope, cyclo[18]carbon

C₁₈ on top of bilayer NaCl on Cu(111)

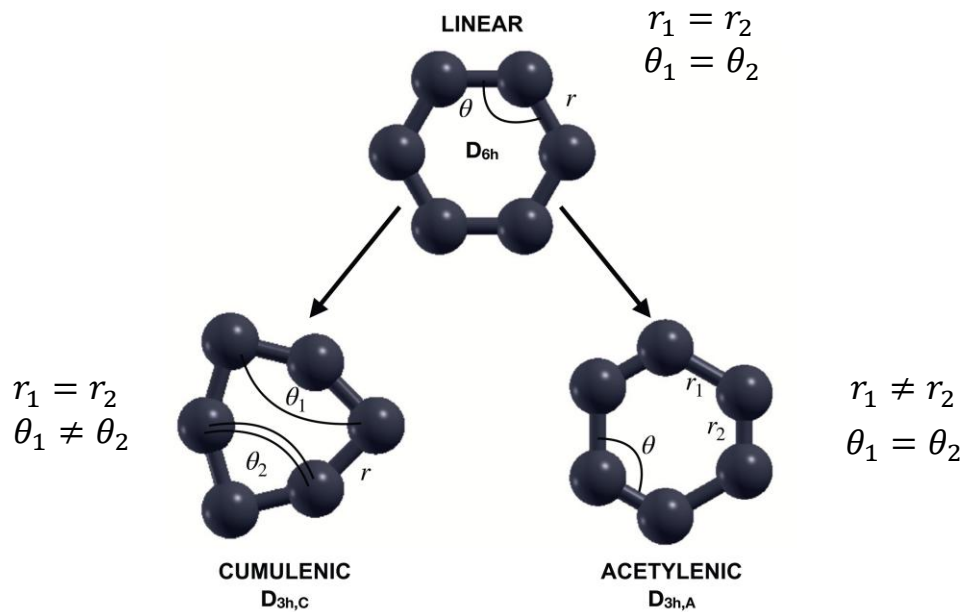
Katharina Kaiser^{1*}, Lorel M. Scriven^{2*}, Fabian Schulz¹, Przemyslaw Gawel^{2†},
Leo Gross^{1†}, Harry L. Anderson^{2†}



C₁₈ : polyynic distorted structure with alternating single and triple bonds.

Cyclon[n]carbons

Monocyclic rings of n carbon atoms: three possible forms (example for n=6)



Accuracy CAMB3LYP functional

ΔE =energy gain with respect to the linear phase

n	Method	Isomer	ΔE [meV/C]	$\langle r \rangle$ [Å]	$\langle \theta \rangle$ [°]	A_r [%]	A_θ [%]
6	QMC [1]	D_{3h} (cumulenic)	65.05	1.33	120.00	0.00	48.4
	CCSD [2]		21.39	/	/	/	/
	CCSD(T) [2]		77.33	/	/	/	//
	CAMB3LYP		128.17	1.31	120.00	0.00	46.42
10	QMC [1]	D_{5h} (cumulenic)	4.7	1.30	144.00	0.00	22.7
	CCSD [2]		37.08	1.31	144.00	0.00	31.44
	CCSD(T) [2]		12.71	1.32	144.00	0.00	29.25
	CAMB3LYP		30.97	1.28	144.00	0.00	27.43
14	QMC [1]	D_{7h} (acetylenic)	16.14	1.29	154.29	7.00	0.00
	CCSD [2]		41.01	1.31	154.29	10.69	0.00
	CAMB3LYP		8.75	1.28	154.29	7.81	0.00
18	QMC [1]	D_{9h} (acetylenic)	22.19	1.29	160.00	7.00	0.00
	CCSD [2]		75.94	1.31	160.00	10.69	0.00
	CAMB3LYP		31.88	1.28	160.00	9.38	0.00

$$\langle r \rangle = \frac{r_1 + r_2}{2}$$

$$\langle \theta \rangle = \frac{\theta_1 + \theta_2}{2}$$

$$A_r = \frac{|r_1 - r_2|}{\langle r \rangle} \times 100$$

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Good agreement with QMC for geometries

[1] T. Torelli, L. Mitas, *Electron correlation in C_{4N} +2 carbon rings: Aromatic versus dimerized structures*, Phys. Rev. Lett. **85** , 1702 (2000)

[2] S. Arulmozhiraja, T. Ohno, *Ccsd calculations on c14, c18, and c22 carbon clusters*, The Journal of Chemical Physics **128** , 114301 (2008)

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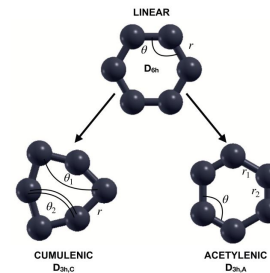
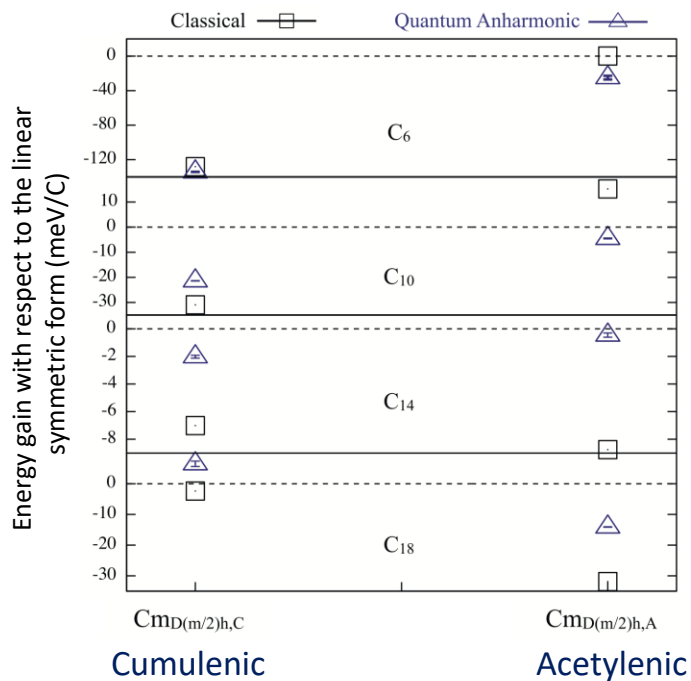
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Quantum anharmonic effects on structural stability

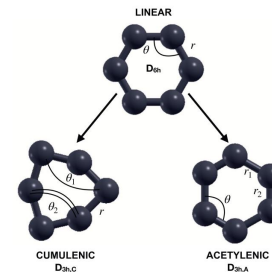
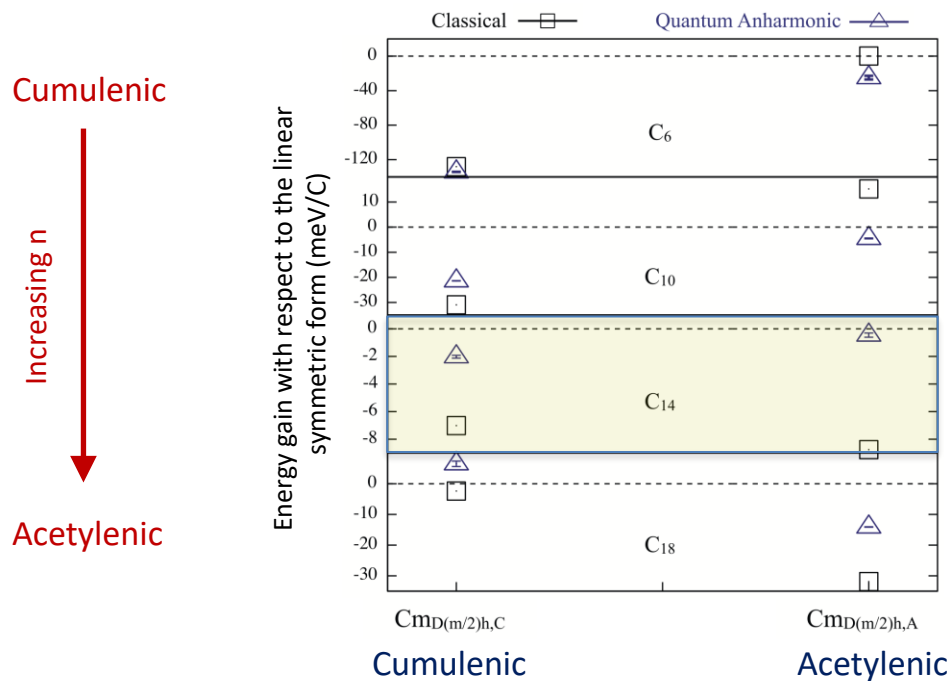
Cumulenic

Increasing n

Acetylenic

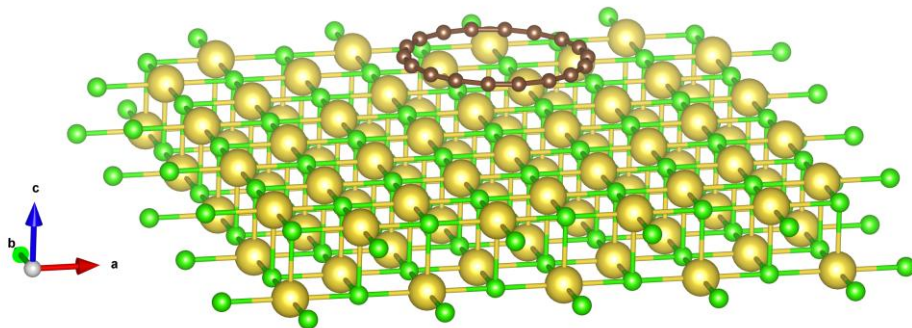


Quantum anharmonic effects on structural stability



C₁₄ is cumulenic only if quantum anharmonic effects are included.

C_{18} on a NaCl bilayer



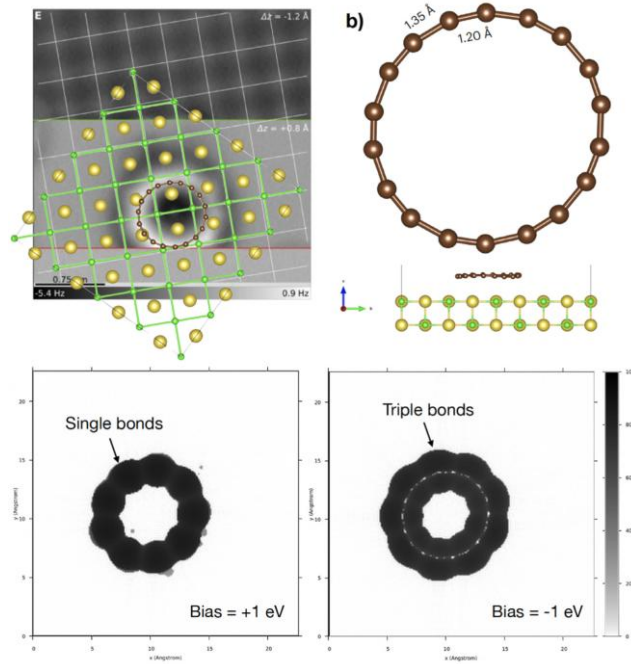
Problem: the substrate affects the screening.

We calculate by using self-consistent GW (evGW) the macroscopic dielectric constant ϵ_{∞} for a double layer NaCl and C₁₈ on top.

$$\alpha^{-1} = \epsilon_{\infty} \quad E_{xc} = \alpha E_x^{HF} + (1 - \alpha) E_x^{GGA} + E_c^{DFT}$$

We obtain $\alpha = 0.75$

C_{18} on a NaCl bilayer

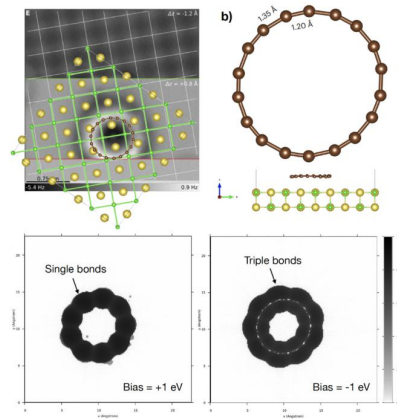


Polyyne distorted structure with alternating single and triple bonds
(in agreement with experiments)

N.B. PBE/LDA/HSE06 functionals fail in this case.

Cyclocarbons - Conclusion

1. Quantum-anharmonicity changes the relative stability of C_{14} cumulenic/acetylenic phases
2. C_{18} is found to be in a polyenyic phase with a bond alternation of $BLA = 0.15 \text{ \AA}$, in agreement with experiments



Stochastic self-consistent harmonic approximation



www.sscha.eu



<https://the-epiq-team.gitlab.io/epiq-site/>

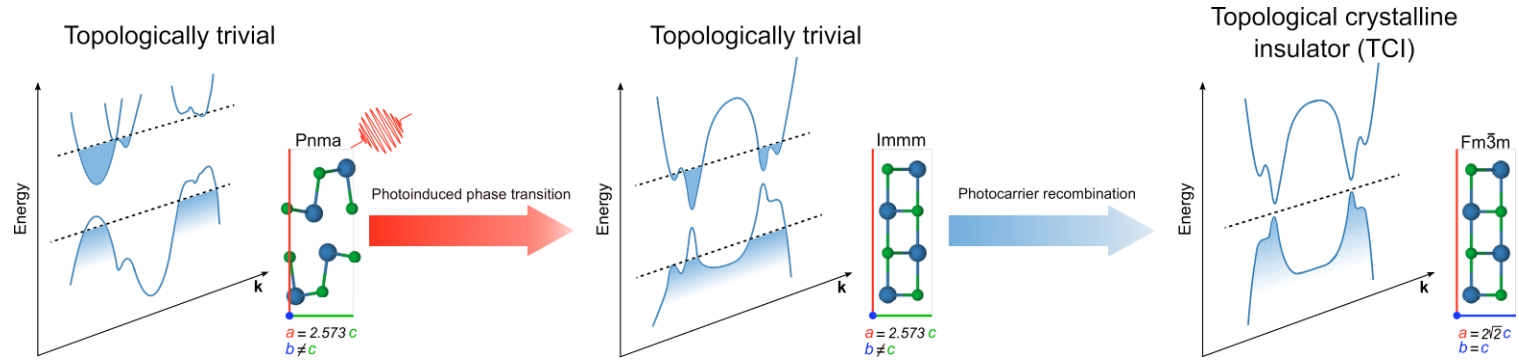


For scientific updates checkout

<https://mattheory.physics.unitn.it>



Non thermal pathway to the topological crystalline phase.



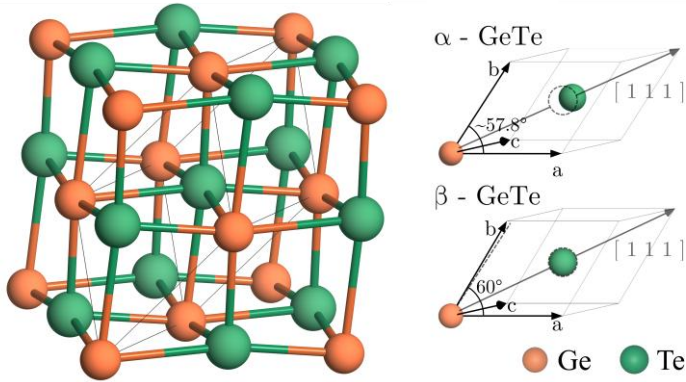
The $Immm$ transient phase spontaneously decay in the TCI SnSe phase.

First order phase transition in photoexcited GeTe

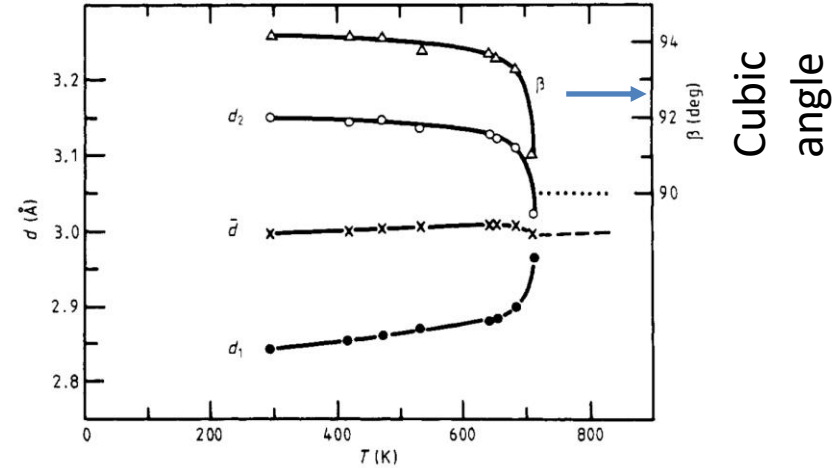
M. Furci, G. Marini and M.C. , Phys. Rev. Lett. **132**, 236101 (2024)

Thermal phase transition

- GeTe high-T phase : Rocksalt (β -phase). Low-T phase Rhombohedral ferroelectric (α -phase) $T_c=670-750$ K
- The distortion is along the (111) direction of the Rocksalt structure.
- Sharp discontinuity of the structural parameter at the transition (displacive phase transition weakly 1st order).

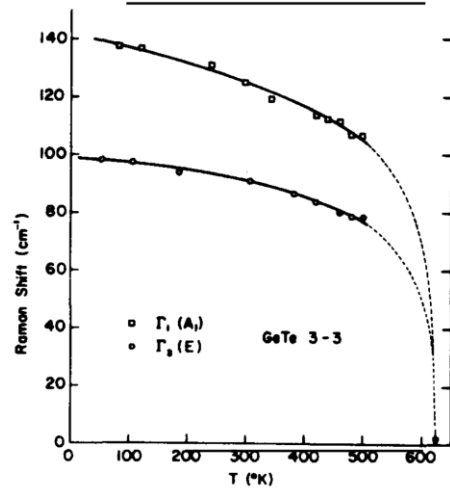


Ge-Te distance

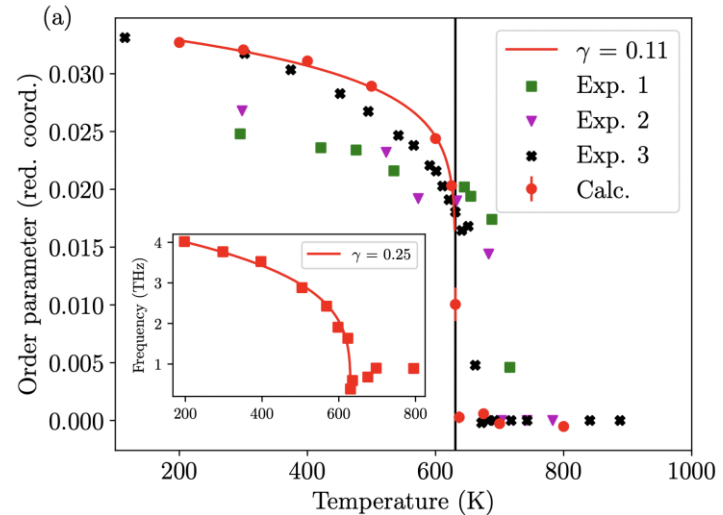


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- Sharp discontinuity of the structural parameter at the transition (displacive phase transition weakly 1st order).
- A phonon softening is detected at the transition, in agreement with MD simulations (2nd order in MD)



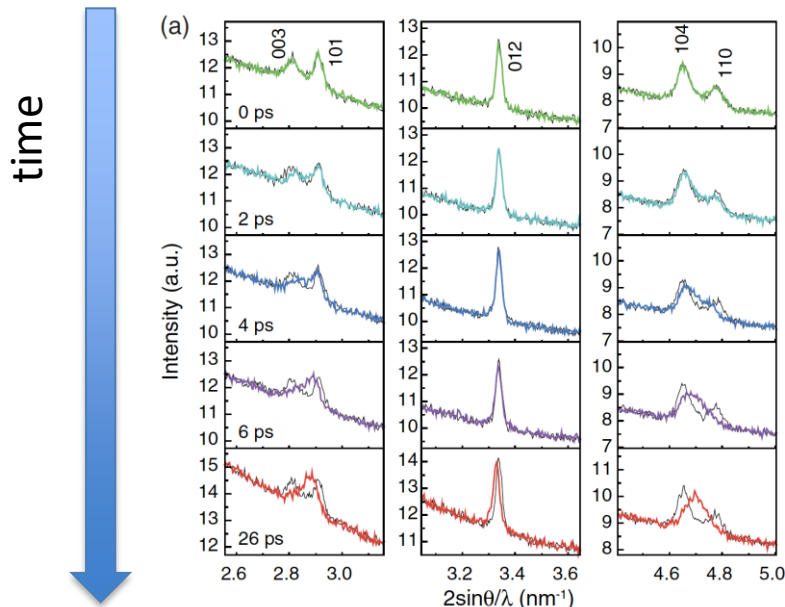
E. F. Steigmeier and G. Harbeke, SSC **8**, 1275 (1970)



Dangic *et al.* PRB **106**, 134113 (2022)

Non-thermal phase transition after photoexcitation with fs pulses.

- Pumping on top of the low-T rhombohedral phase leads to a sort of rocksalt cubic phase



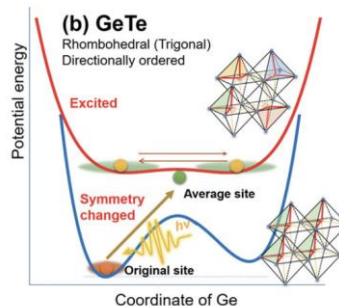
E. Matsubara *et al.* PRL **117**, 135501 (2016)

- Ultrafast XRD: Rhombohedral peak slightly suppressed (but still present after 26 ps)

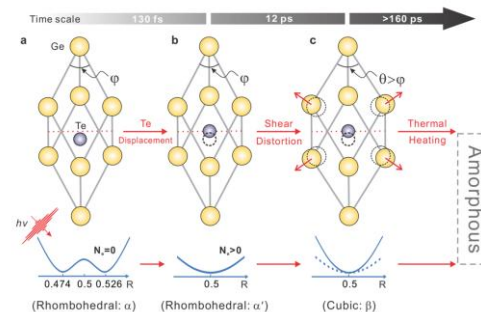
Cubic peaks emerging after 4-6 ps.

- Two controversial interpretations for the persistence of R-peaks

Order-disorder

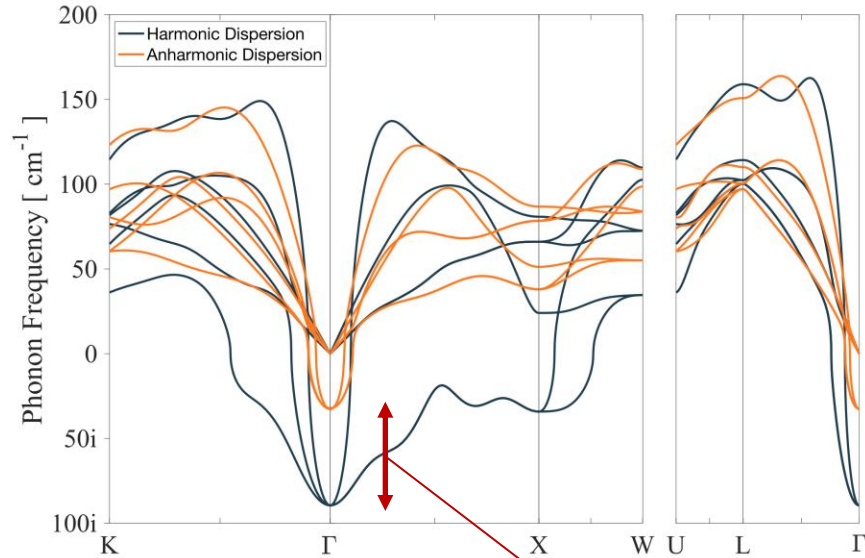


Hidden Transient R-structure



Thermal phase transition

- As a preliminary step we show that the SSCHA is able to reproduce the phonon instability of cubic GeTe at T=300 K

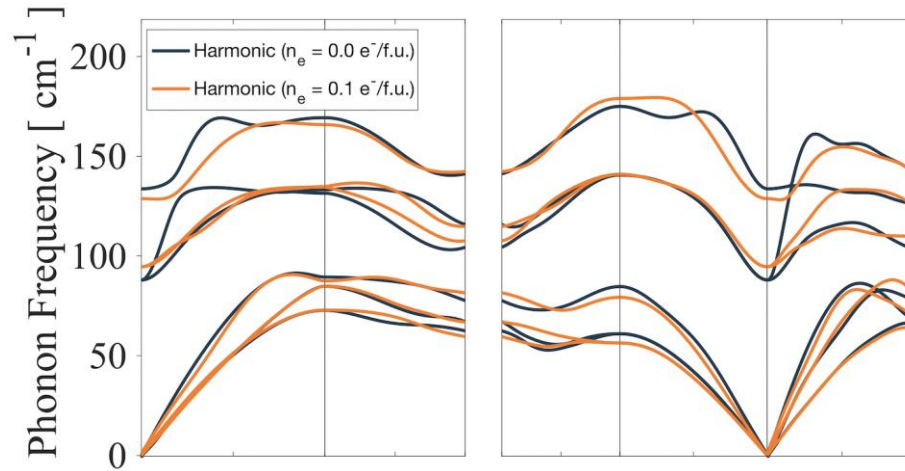


CAVEAT: LO/TO missing

Anharmonic shift

Effect of photoexcitation on the rhombohedral phase

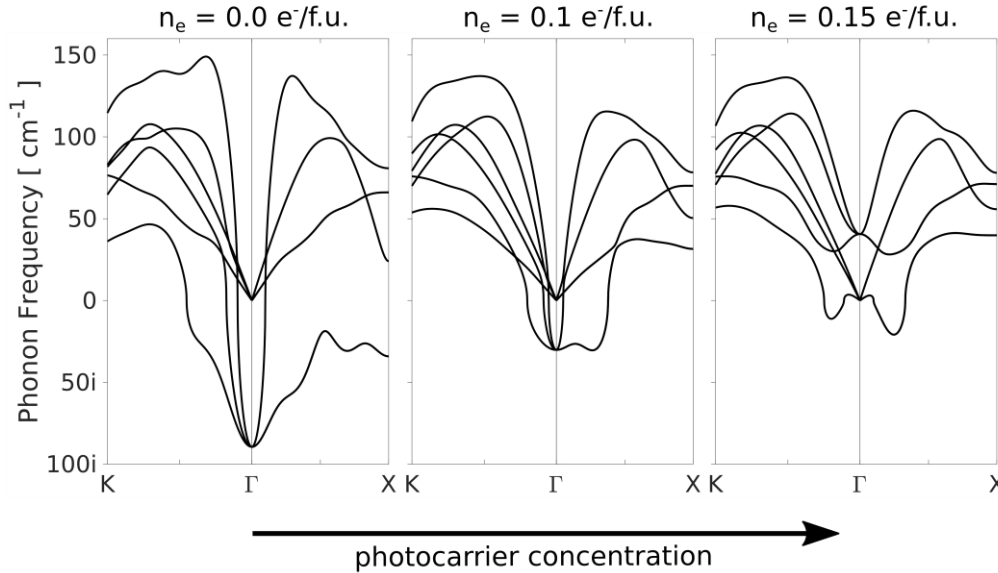
- At $n=0.1$ photoexcited carriers per unit cell the rhombohedral phase do not display any instability



No second-order structural instability

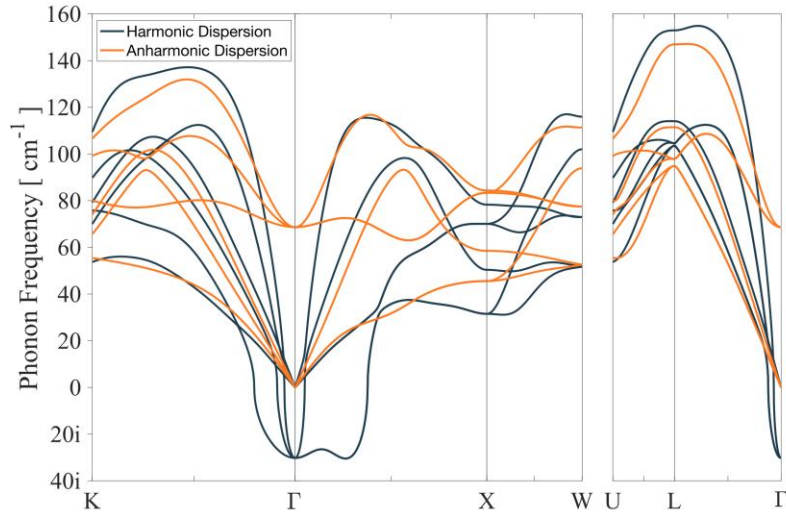
Effect of photoexcitation on the cubic phase

- At the harmonic level the zone center structural instability is reduced by photoexcitation.



Effect of photoexcitation on the cubic phase

- At the harmonic level the zone center structural instability is reduced by photoexcitation.
- Anharmonic effects completely remove the instability of the photoexcited phase at $n=0.1$



Both structures are dynamically stable after photoexcitation .
First order transition ?

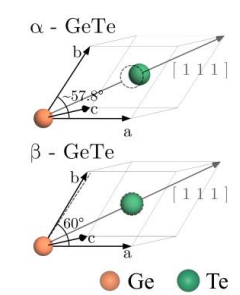
Effect of photoexcitation on the cubic phase

- At the harmonic level the zone center structural instability is reduced by photoexcitation.
- Anharmonic effects completely remove the instability of the photoexcited phase at $n=0.1$
- Energetics

Ground state

Total energy difference

$$\Delta E = E_{\alpha} - E_{\beta} = 29 \text{ meV/f.u.}$$



Quantum free energy difference (T=300K)

$$\Delta F = F_{\alpha} - F_{\beta} = 18 \text{ meV/f.u.}$$

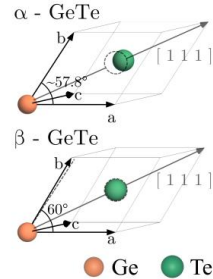
Effect of photoexcitation on the cubic phase

- At the harmonic level the zone center structural instability is reduced by photoexcitation.
- Anharmonic effects completely remove the instability of the photoexcited phase at $n=0.1$
- Energetics

Ground state

Total energy difference

$$\Delta E = E_{\alpha} - E_{\beta} = 29 \text{ meV/f.u.}$$



Excited state $n=0.1$

$$\Delta E = E_{\alpha} - E_{\beta} = 15.9 \text{ meV/f.u.}$$

Quantum free energy difference (T=300K)

$$\Delta F = F_{\alpha} - F_{\beta} = 18 \text{ meV/f.u.}$$

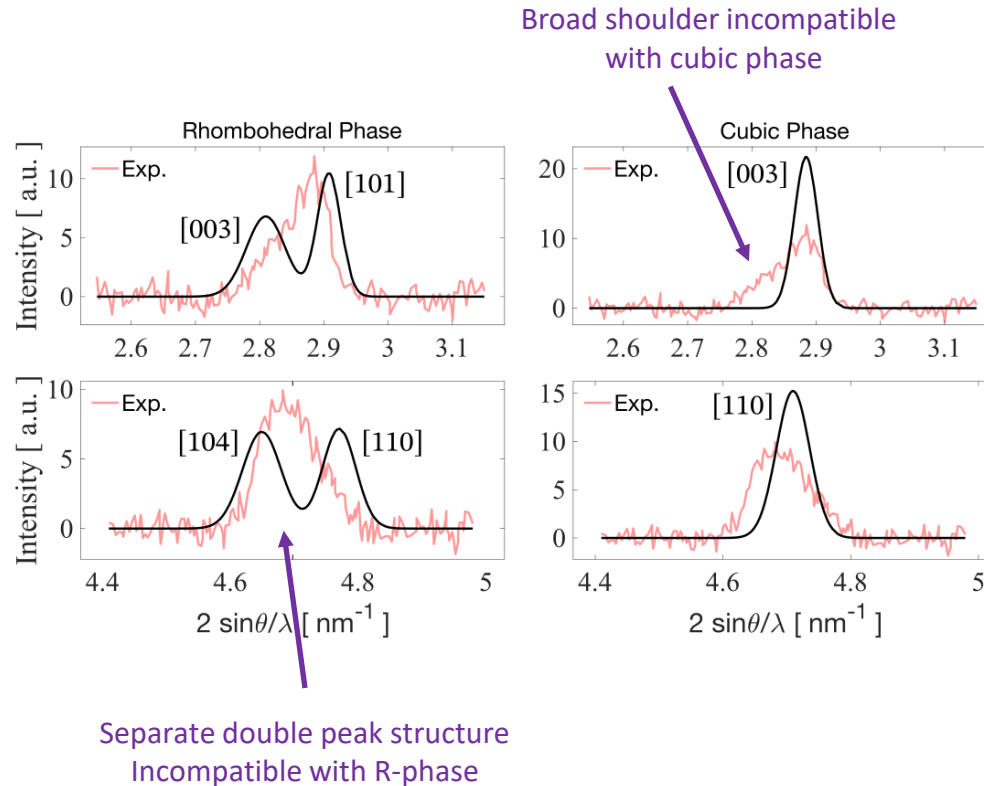
$$\Delta F = F_{\alpha} - F_{\beta} = 2.36 \text{ meV/f.u.}$$

Electronic excitation makes the two structures degenerate and dynamically stable!

Can the coexistence between the two phases explain ultrafast XRD data ?

E. Matsubara *et al.* PRL **117**, 135501 (2016)

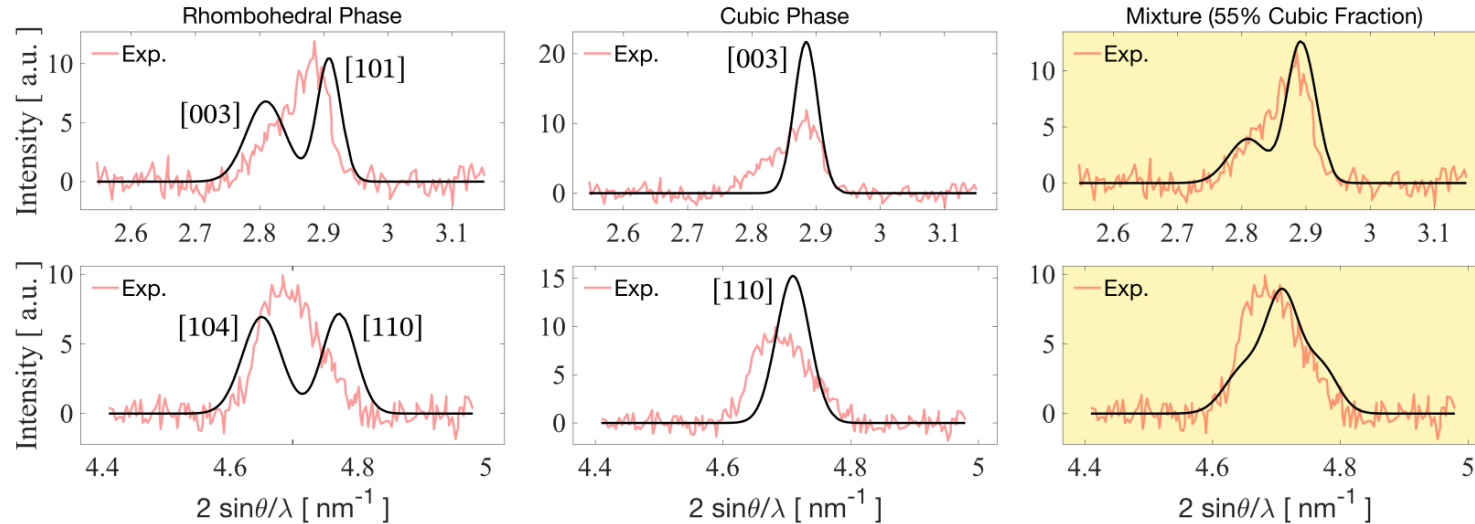
- XRD data 6 ps after the transition



Can the coexistence between the two phases explain ultrafast XRD data ?

E. Matsubara *et al.* PRL **117**, 135501 (2016)

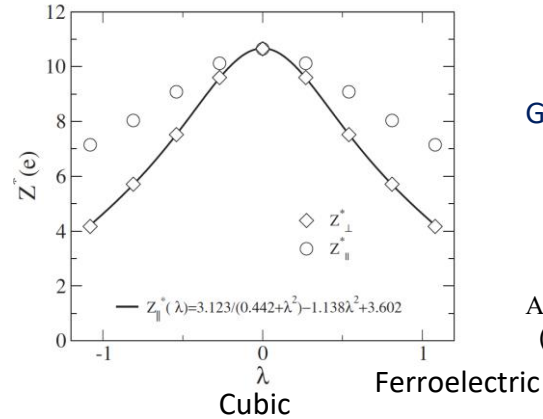
- XRD data 6 ps after the transition



XRD Data are more consistent with a 55% phase coexistence.

Mechanism for the first order transition

- In the ground state the reduced ionicity and larger gap of the rhombohedral phase stabilize the distortion.

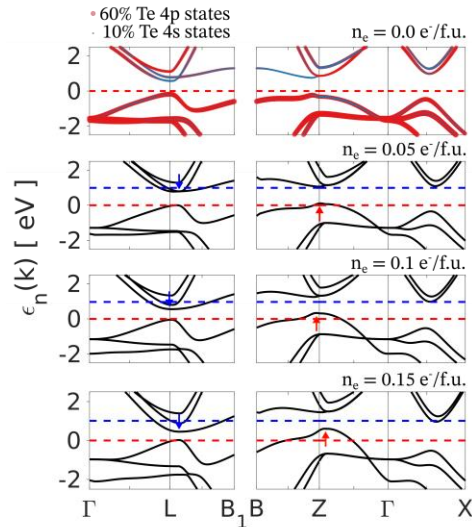


Ge Born effective charges

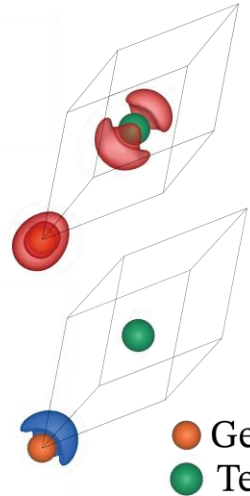
Alshaf *et al.* PRB **78**, 205203
(2008)

Mechanism for the first order transition

- In the ground state the reduced ionicity and larger gap of the rhombohedral phase stabilize the distortion.
- Photoexcitation reduces ionicity (Te $\rightarrow$ Ge electron transfer) but stabilizes the cubic structure (contradiction?).



Rhombohedral phase

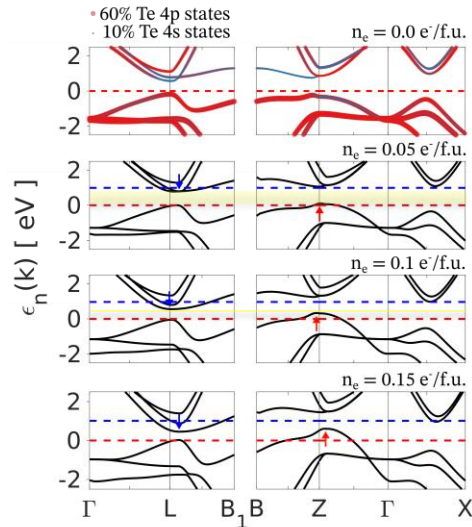


Valence band top
charge density

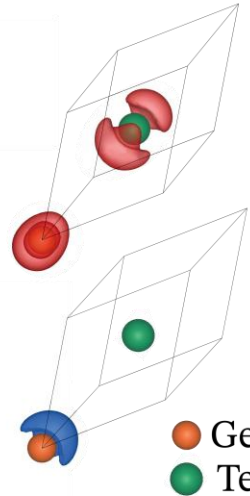
Conduction band bottom
charge density

Mechanism for the first order transition

- In the ground state the reduced ionicity and larger gap of the rhombohedral phase stabilize the distortion.
- Photoexcitation reduces ionicity (Te $\rightarrow$ Ge electron transfer) but stabilizes the cubic structure (contradiction?).
- The high-density electron-hole plasma closes the gap and reduces the energy difference.



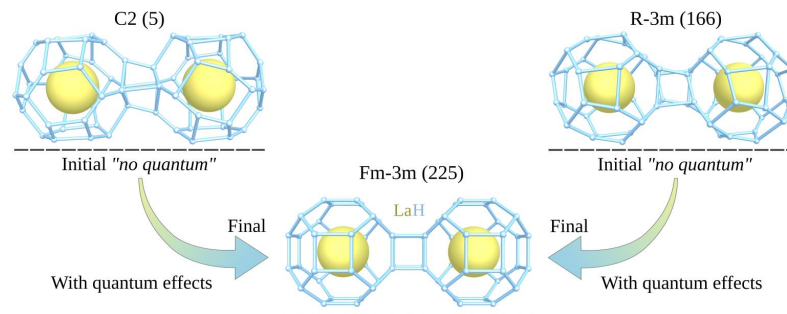
Rhombohedral phase



Valence band top
charge density

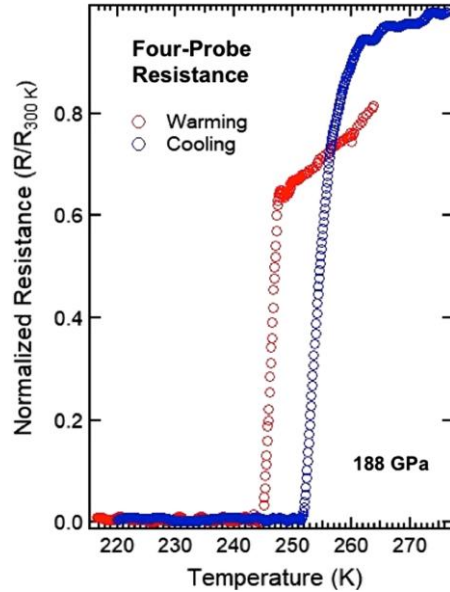
Conduction band bottom
charge density

LaH₁₀

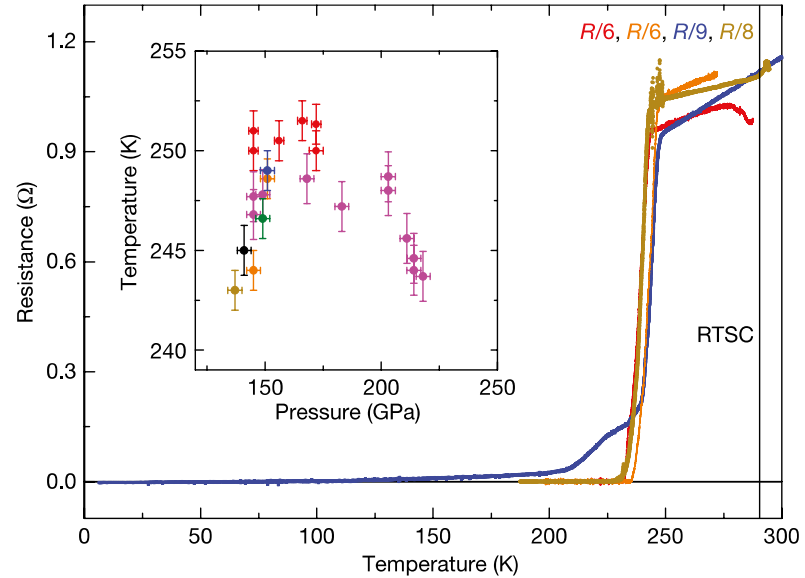


Errea *et al.* Nature **578**, 66–69(2020)

Superconductivity at $T_c=250$ K in LaH_{10}

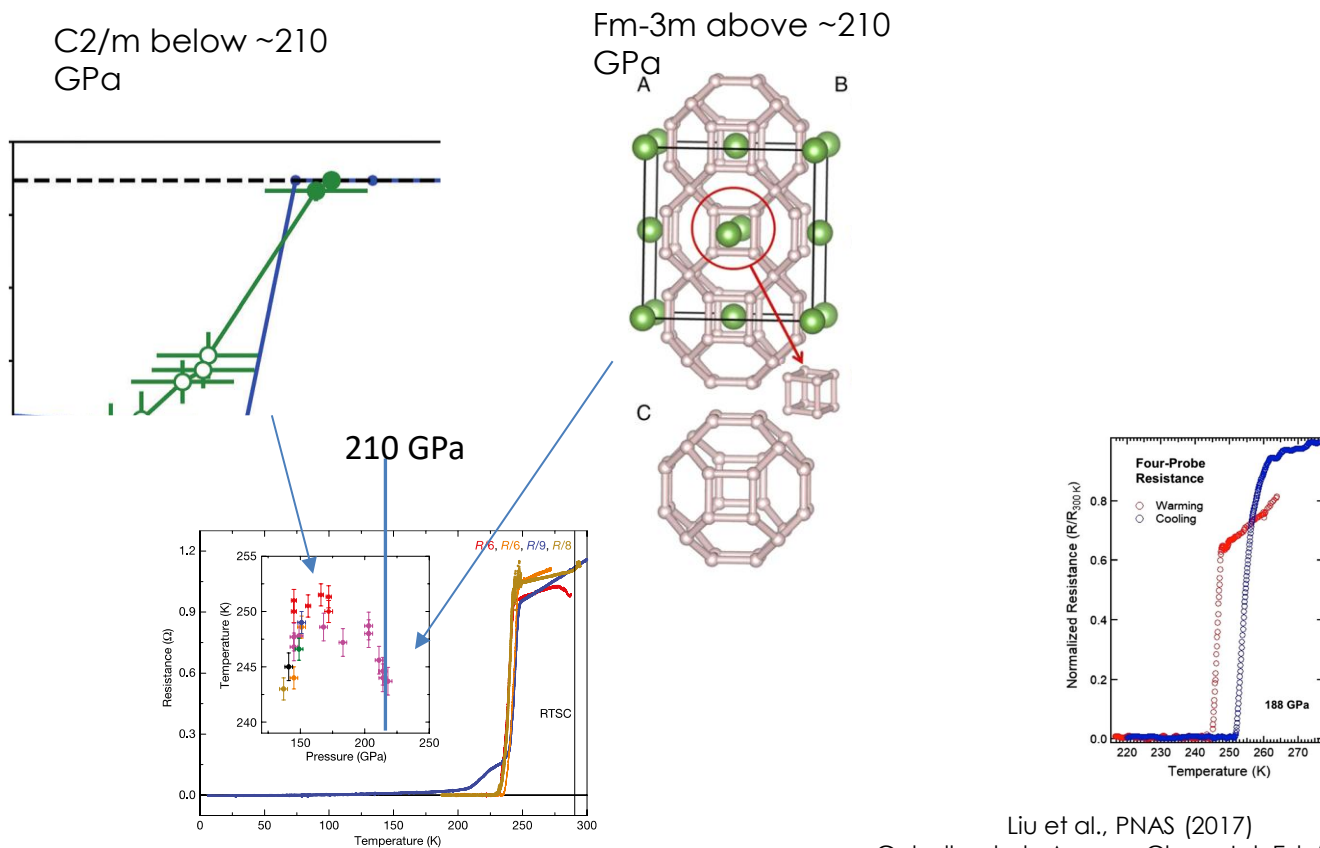


Somayazulu et al., PRL(2019)



Drozdov et al., Nature (2019)

Without quantum effects: superconductivity is due $C2/m$



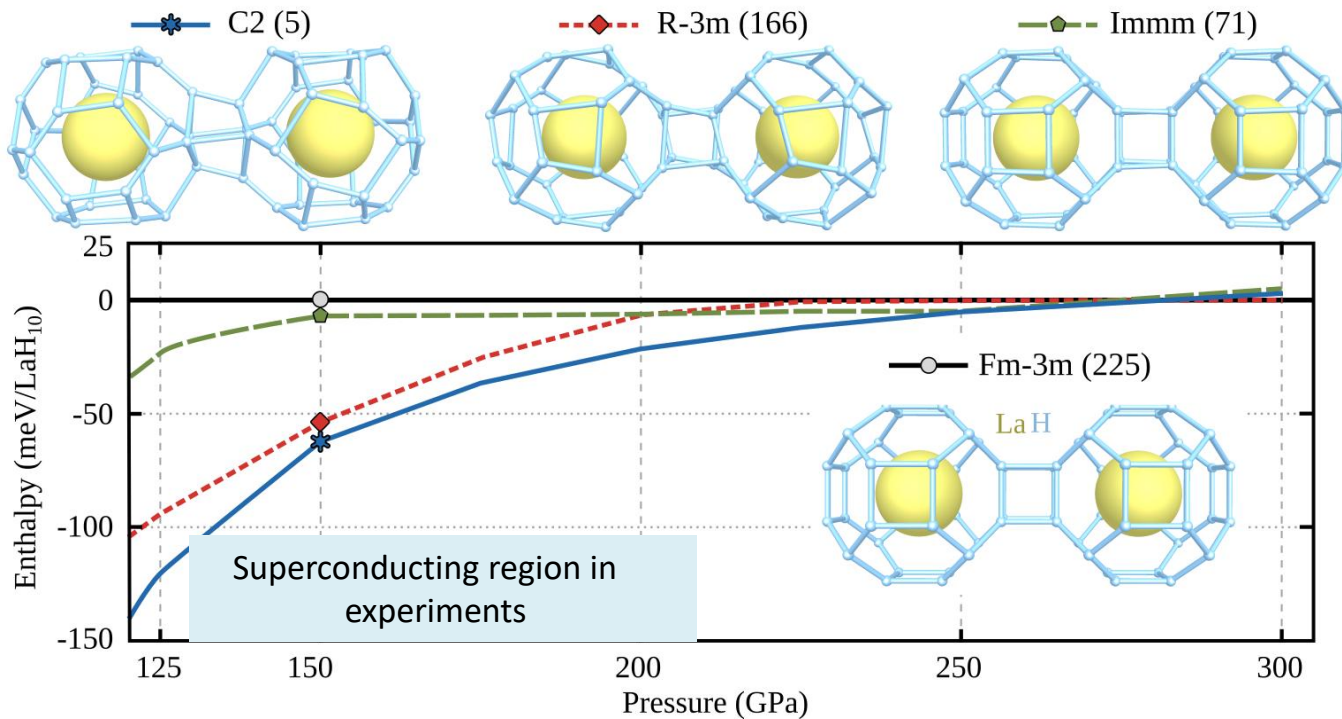
Liu et al., PNAS (2017)

Geballe et al., Angew. Chem. Int. Ed. (2018)

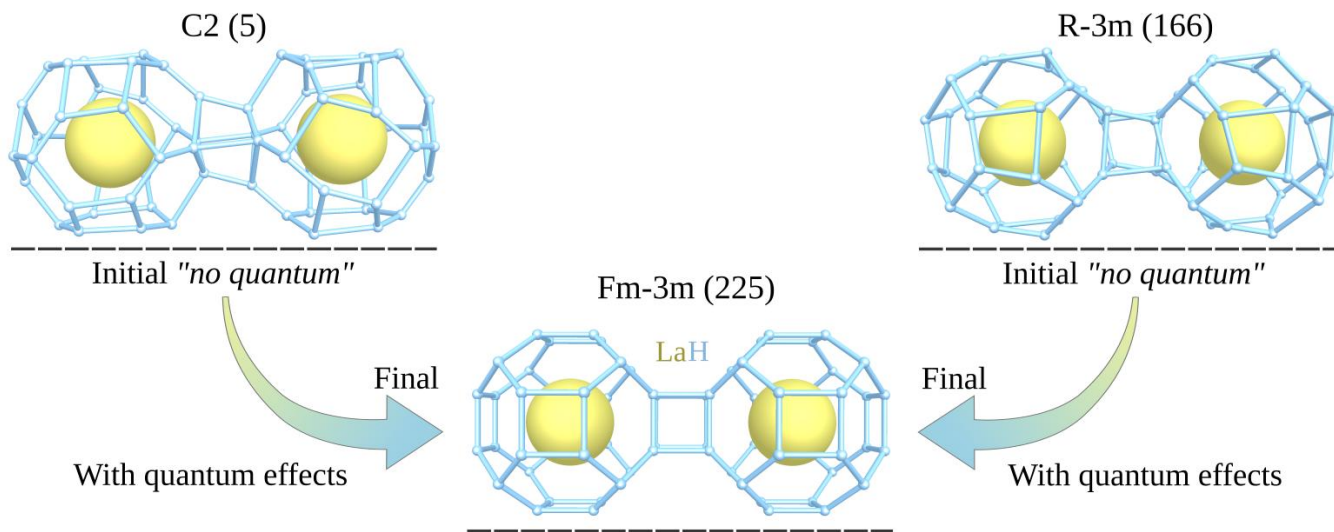
Somayazulu et al., PRL(2019)

Drozdov et al. Nature (2019)

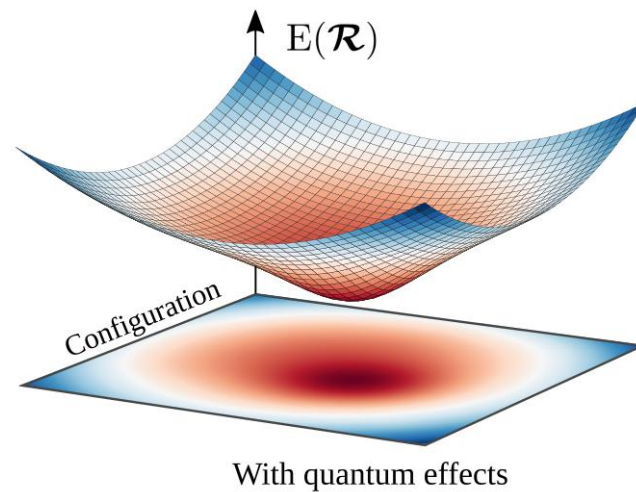
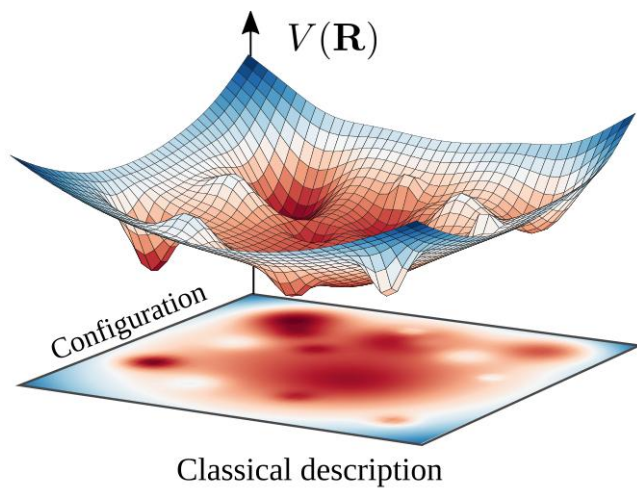
Without quantum effects: superconductivity is due $C2/m$



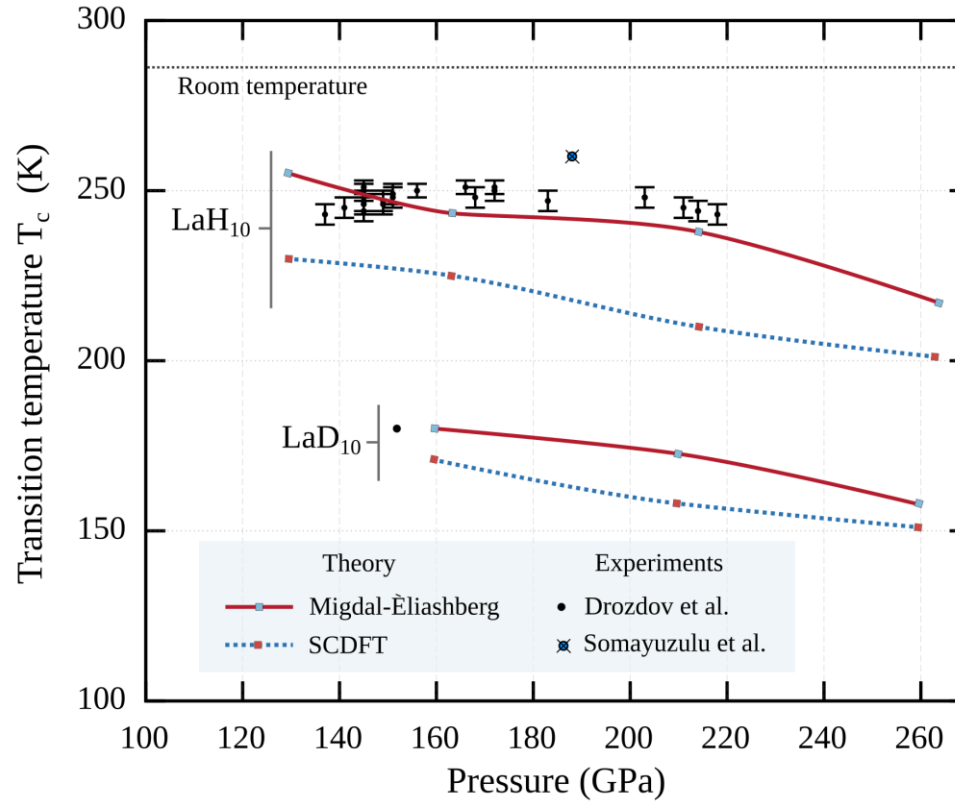
Quantum Structural optimization (with Quantum pressure !)



Quantum energy landscape



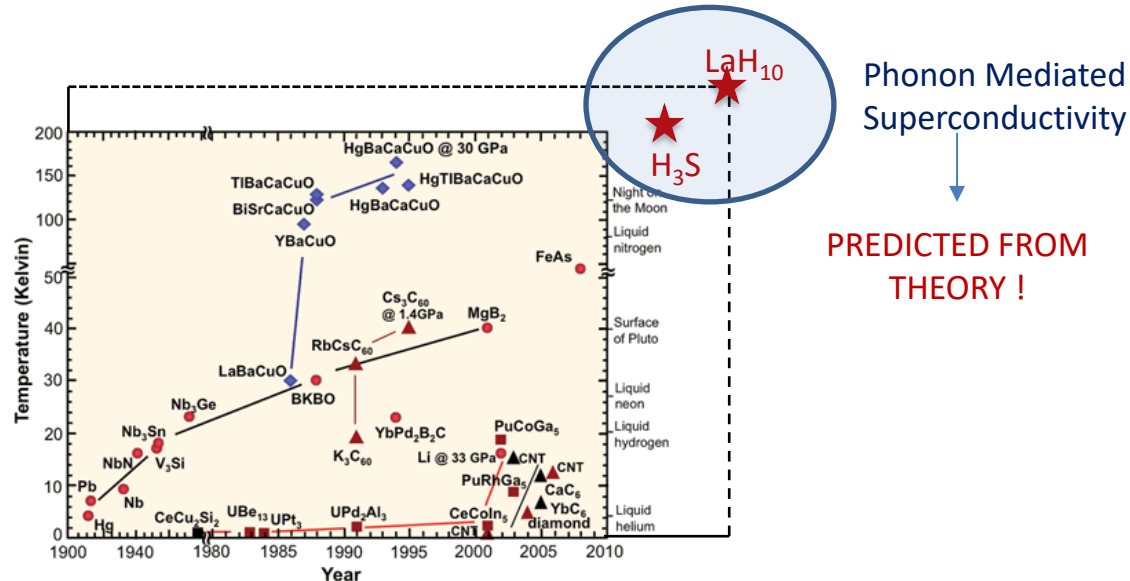
Calculated T_c in agreement with experiments in all the pressure range !



Critical superconducting temperature calculated with manybody + ab initio

Conclusion

- ➡ Proton quantum effects reshape the phase diagram of H-rich solids
- ➡ The metallization of H occurs at 360 GPa via indirect band-gap closure
- ➡ Above 360 GPa Hydrogen is in a dark phase, invisible to optics up to 460 GPa
- ➡ Room temperature Superconductivity hard to achieve in H but possible in superhydrides

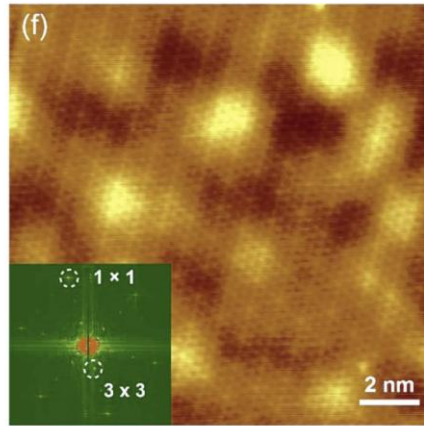


Bulk and single (tri)layer 1HNbS₂

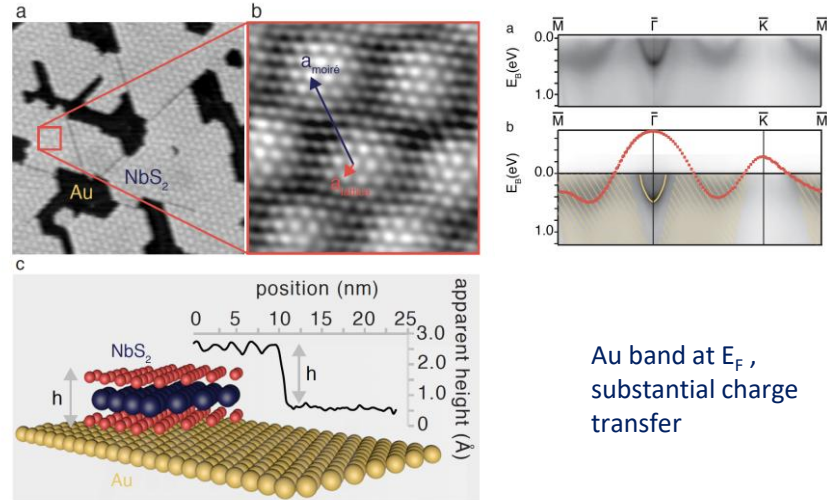
R Bianco, I Errea, L Monacelli, M Calandra, F Mauri, Nano letters 19, 3098 (2019)

Single layer 1H NbS₂ on different substrates

- ➔ Single (tri)layer MBE grown samples of NbS₂ on graphene/6HSiC(0001) and on Au(111)
- ➔ 3x3 Charge density wave appearing on graphene/6HSiC(0001) and absent on Au(111)



FFT of STM

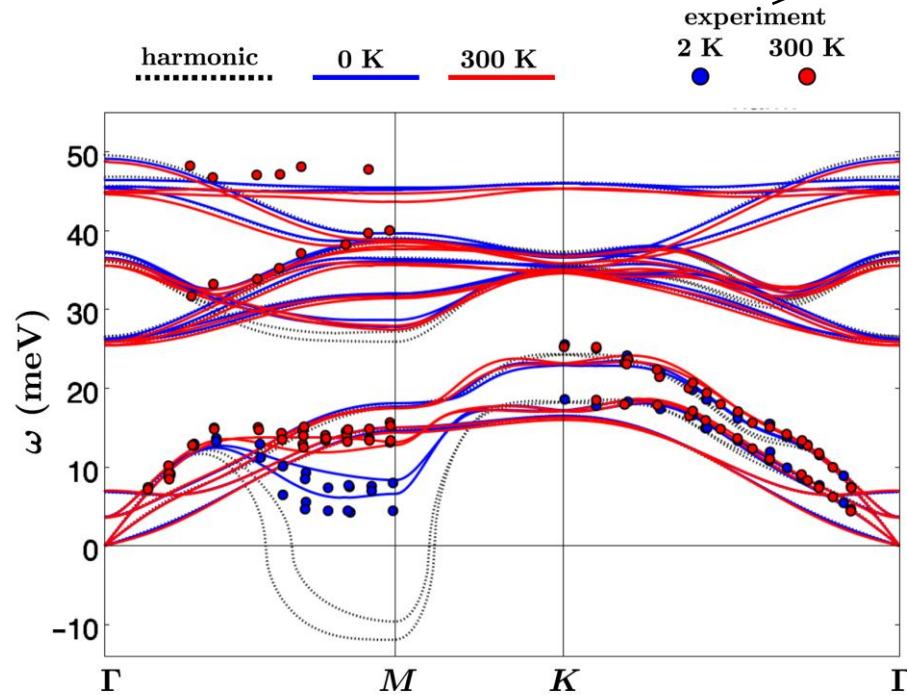


Au band at E_F ,
substantial charge
transfer

Strong sensitivity of the CDW to environmental conditions.

Bulk 2HNbS_2 – Anharmonic suppression of CDW

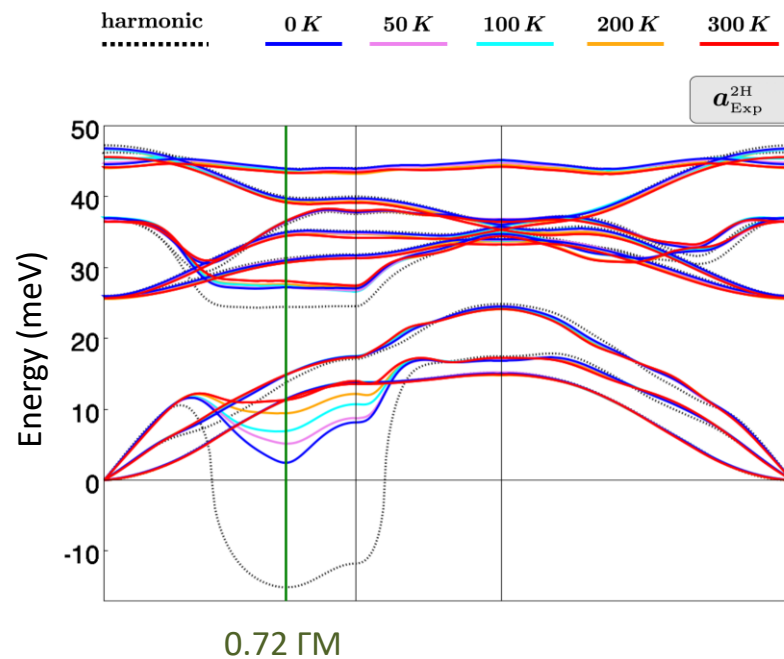
- Harmonic calculations suggest 2x2 CDW, while no CDW is present in bulk.
- Strongly temperature dependent IXS phonon spectra. Leroux *et al.*, Phys. Rev. B 86, 155125 (2012)



Anharmonicity explains temperature dependence of phonon spectra.

Single layer 1HNbS₂

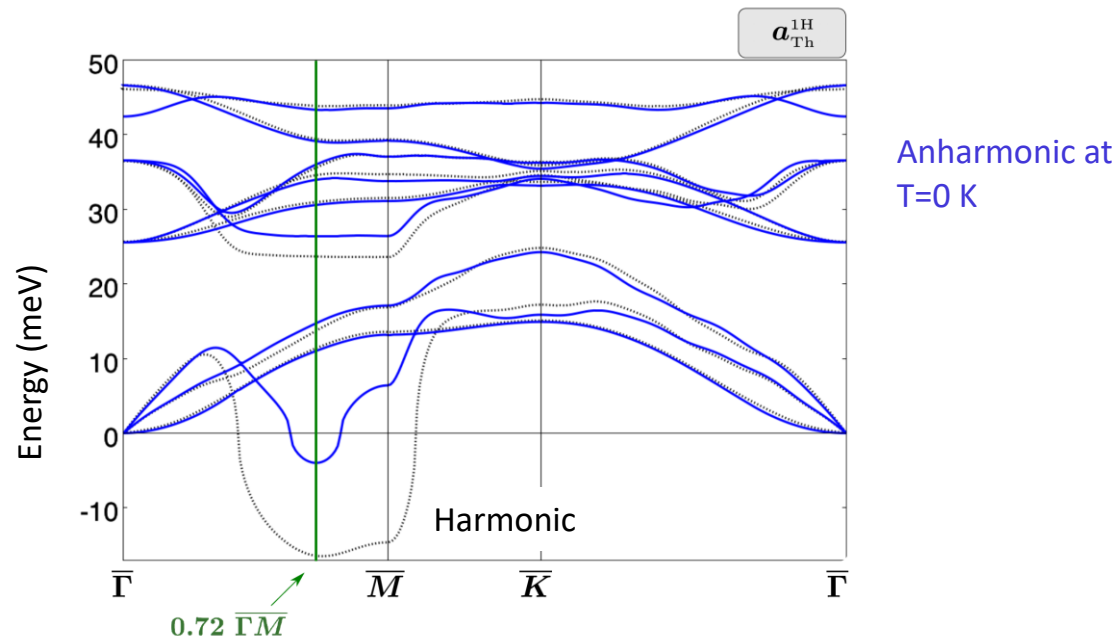
- ➔ By using the experimental lattice parameter the system is at the verge of a CDW.
- ➔ Soft-mode 20% softer than in the bulk



Sensitivity to strain ?

Single layer 1HNbS₂

- The theoretical anharmonic lattice parameter is 0.5% expanded with respect to the bulk
- 3x3 charge density wave sets in.

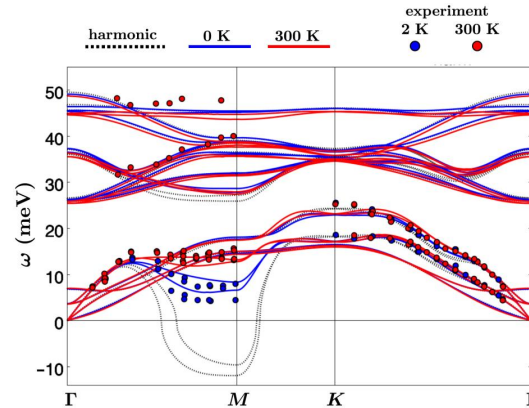


Extreme sensitivity to environmental conditions.

Quantum enhancement of CDW in the 2D limit.

Conclusions (1HNbS₂)

- Crucial role of anharmonicity in determining both the occurrence of CDW (see 1HNbS₂) and the magnitude of T_{CDW} in dichalcogenides.



- In 1HNbS₂ is at the verge of the CDW transition and small environmental perturbations can suppress it or enhance it.

JS Zhou, L Monacelli, R Bianco, I Errea, F Mauri, M Calandra Nano Letters 20, 4809 (2020)

JS Zhou, R Bianco, L Monacelli, I Errea, F Mauri, M Calandra, 2D Materials 7, 045032 (2020)

R Bianco, I Errea, L Monacelli, M Calandra, F Mauri, Nano letters 19, 3098 (2019)

Stochastic self-consistent harmonic approximation



www.sscha.eu



<https://the-epiq-team.gitlab.io/epiq-site/>



For scientific updates checkout

<https://mattheory.physics.unitn.it>



What are the typical fluences and the number of photoexcited carriers ?

- Single-shot damage threshold depends on the materials and, even more, on the experimental setting.

SYSTEM	Fluence used (mJ/cm ²)	Photocarriers/cell	Damage threshold (mJ/cm ²)
VO ₂ [1,2]	25	0.31	65
GeTe [3,4]	7-20	0.1-0.36	35-50 (amorphysation)
SnSe [4,5]	2.2 (in vacuum)	0.3	>3 (caveat: T=300, inefficient heat diss.)
SnSe [6]	13.2 (on top of diamond)	1.8	

Sample damage is mainly related to difficulties in heat dissipation.

Different set up show very different damage threshold (see later)

[1] A. Cavalleri *et al.* PRL 87, 237401 (2001).

[2] S. Wall *et al.* Science 362, 572 (2018)

[3] J. Hu *et al.* ACS Nano 9, 6728 (2015)

[4] E. Matsubara *et al.* PRL **117**, 135501 (2016)

[5] Huang *et al.* PRX 12, 011029 (2022)

[6] B. J. Dringoli *et al.* arXiv:2212.12498